

# FROM MARGINAL MATCHING TO SAMPLER COMPATIBILITY: AUDITING INDEPENDENT LATENT RECOMBINATION

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## ABSTRACT

During training, split-latent generators often reconstruct an observation from content and style codes encoded from that same observation; at generation, their decoder inputs may instead be sampled independently. We formalize this train-generation shift as *sampler compatibility* between the encoded conditional joint  $Q_y = q(z_c, z_s | y)$  and the declared sampling law  $P_y = p(z_c | y)p(z_s)$ . An exact expected-conditional KL identity decomposes incompatibility into style-prior mismatch, condition information in style, content-prior mismatch, and within-condition code dependence. We pair a direct joint test with clause-specific diagnostics, decoder interventions, and task metrics, thereby separating latent compatibility, decoded transport, and generation competence. An exact-marginal counterexample and controlled audits on image and known-factor data show that aggregate matching can coexist with strong leakage and decoded style-donor control. In three fully traceable matched MNIST pairs, a *factorized-contract training* (FACT) intervention reduces the selected joint-MMD diagnostic by  $38.9 \pm 3.6\%$  and style-probe accuracy from 76.9% to 11.1%, but lowers generation accuracy by  $9.9 \pm 2.8$  points; a post-specified energy distance does not improve in every pair. Thus aggregate matching is only one clause of independent recombination, and improving a chosen compatibility diagnostic need not improve generation competence.

## 1 INTRODUCTION

Many split-latent generators are trained and sampled in different ways. During reconstruction, content and style codes come from the same observation. During generation, the two codes are drawn or recombined independently. If the decoder has learned dependencies that occur only in paired codes, generation queries it under a latent distribution it did not observe during reconstruction.

Let  $Q_y = q(z_c, z_s | y)$  denote the encoded pair conditional on the requested condition  $y$ , and let  $P_y = p(z_c | y)p(z_s)$  denote the independently sampled decoder-input law. We call  $Q_y = P_y$  *sampler compatibility*. Throughout, “content” means the code intended to carry the requested condition and “style” means the designated residual code; these are operational roles, not universal perceptual definitions. The contract applies only when a model uses independent recombination. A generator that models and samples  $Q_y$  directly makes no such distributional substitution.

Aggregate prior matching checks only one projection of this joint claim. It may make  $q(z_s)$  close to a Gaussian without ensuring that style is invariant to  $y$ , that encoded content follows its sampling prior, or that the two codes are independent within each condition. Good reconstruction and a small global prior discrepancy can therefore coexist with  $Q_y \neq P_y$ .

A failure of sampler compatibility is not, by itself, a quality verdict. A decoder can be insensitive to the mismatched directions, while matched latent laws do not guarantee realistic or task-correct outputs. We consequently distinguish three questions: latent *compatibility* ( $Q_y = P_y$ ), *transport compatibility* ( $G_{\#}Q_y = G_{\#}P_y$ ), and *generation competence* relative to the real conditional data law and intended task. MNIST swaps show a mismatch that the decoder actively uses; a separate SCFlow boundary case shows why endpoint and task evidence must still be interpreted in the native model space.

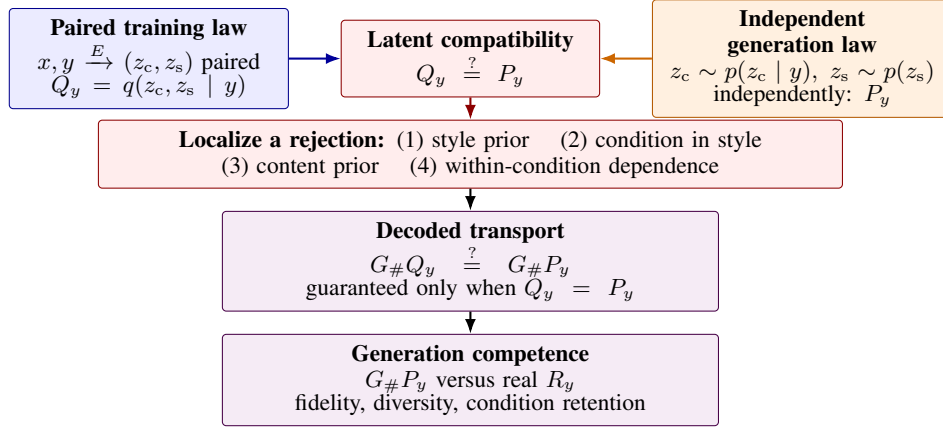


Figure 1: The audit starts from the model’s actual training and generation laws. Latent equality implies decoded equality for a fixed decoder; neither implies agreement with real data, so transport and competence require separate evidence.

Our technical contribution is not a new KL rule: it is the sampler-specific reference joint and the resulting audit target. The decomposition turns familiar marginal diagnostics into necessary but individually insufficient clauses, adds content-prior and conditional block-dependence requirements, and separates latent failure from decoded consequence. We ask whether aggregate matching can conceal incompatibility, whether the decoder uses that mismatch, and what changes when the diagnosed clauses are optimized directly.

Our contributions are:

1. We formulate the declared generation law as a falsifiable joint contract, derive an exact four-term KL decomposition, and distinguish latent compatibility, decoded transport, and task competence.
2. We operationalize a direct joint equality test followed by clause-specific attribution and decoder/task interventions; the diagnostics test zero-sets rather than claim to estimate the four KL magnitudes.
3. Across exact constructions, controlled image models, Shapes3D, and a released-checkpoint boundary case, we show that the audit distinguishes null-like and consequential failures. A fully traceable matched intervention then exposes both diagnostic gains and their generation cost.

## 2 RELATED WORK

**Aggregate matching and latent decomposition.** Aggregate-prior metrics examine one marginal; leakage probes examine whether a designated representation predicts a condition; factor scores describe selected cross-loadings. Each is useful, but none alone tests the conditional joint substituted at generation. This distinction connects several familiar objectives. VAEs (Kingma & Welling, 2013) penalize per-example KL, whereas WAEs (Tolstikhin et al., 2017) match aggregated posteriors with objectives such as MMD (Gretton et al., 2012). The averaged VAE KL separates information capacity from aggregate-prior mismatch (Hoffman & Johnson, 2016);  $\beta$ -VAE,  $\beta$ -TCVAE, FactorVAE, and DIP-VAE further target capacity, unconditional total correlation, or aggregate moments (Higgins et al., 2017; Chen et al., 2018; Kim & Mnih, 2018; Kumar et al., 2018). Our reference is instead the conditional product law induced by the declared sampler, which additionally requires a content prior and block independence given  $y$ . Aggregate matching is therefore not incorrect; it certifies one marginal of a larger joint claim.

**Learning content–style invariance and preventing leakage.** Split-latent models use grouped observations, labels, paired supervision, or latent optimization to separate specified and residual

variation (Mathieu et al., 2016; Bouchacourt et al., 2018; Jha et al., 2018; Ilse et al., 2020; Gabbay & Hoshen, 2020). Leakage filtering directly probes class information in style before recombination (Ridgeway & Mozer, 2018); fair representation, adversarial, Fader, and information-based objectives suppress designated nuisance signals (Louizos et al., 2015; Ganin & Lempitsky, 2015; Lample et al., 2017; Moyer et al., 2018). Recent diffusion and autoregressive systems explicitly suppress visual-prompt or style-code content leakage, including StyleKeeper and CSD-VAR (Jeong et al., 2025; Nguyen et al., 2025), as well as feature masking and variance-invariance approaches (Zhu et al., 2025; Wu et al., 2025). These methods target leakage or synthesis quality. Here condition information in style is one possible source of  $Q_y \neq P_y$ ; it can vanish while content-prior mismatch or conditional code dependence remains. More broadly, unsupervised factor identification requires assumptions on the model or data (Locatello et al., 2019).

**Disentanglement versus editability.** Recent theory formalizes information-theoretic content-style separation and editability as complementary properties, and gives identifiability and finite-sample results for diffusion models under structural assumptions (Wang et al., 2025). We do not prove factor identifiability or the existence of an editable decoder. Instead, for a fixed trained encoder, decoder, and declared sampler, we test whether the paired decoder-input law is the law used for independent recombination. Our transport check is therefore sampler- and decoder-specific rather than an alternative definition of editability.

**Recombination and its audit.** MUNIT and DRIT combine content with sampled style (Huang et al., 2018; Lee et al., 2018); recent methods use variance patterns or invertible merging to obtain content-style structure (Wu et al., 2025; Ma et al., 2025). These methods evaluate decoded quality and diversity under recombination. Our complementary question is whether the encoded conditional joint equals the particular input law used by that recombination, which clauses differ if it does not, and whether the difference survives decoding.

**Known-factor evaluation.** Class labels are only one semantic partition. Controlled datasets such as Shapes3D expose shape, color, scale, and orientation factors and have been used to evaluate whether representation coordinates track known generative variables (Higgins et al., 2017; Locatello et al., 2019; Google DeepMind, 2018). We use them differently: a designated content/style split is tested in both directions, with family-wise corrected dependence tests and pixel-space swap evaluation.

### 3 FACTORIZED SAMPLING AND SAMPLER COMPATIBILITY

We now formalize the distributional replacement performed by factorized sampling and derive the conditions under which it disappears.

Let  $q(y)$  denote the empirical distribution of an externally specified condition and let an encoder induce  $q(z_c, z_s | y)$ . We condition on the requested  $y$  throughout; matching a separate sampling law  $p(y)$  is outside this contract and would add  $\text{KL}(q(y) || p(y))$  to a full-joint comparison. A factorized class-conditional sampler draws

$$z_c \sim p(z_c | y), \quad z_s \sim p(z_s), \quad (1)$$

independently. In our case  $p(z_s) = \mathcal{N}(0, I)$ . Many aggregate objectives constrain only

$$\mathcal{L}_{\text{style}} = D(q(z_s), p(z_s)), \quad q(z_s) = \sum_y q(y) q(z_s | y), \quad (2)$$

For the remainder, write

$$Q_y := q(z_c, z_s | y), \quad P_y := p(z_c | y)p(z_s), \quad \boxed{\text{sampler compatibility: } Q_y = P_y \text{ } q(y)\text{-a.e.}} \quad (3)$$

Equation 1 covers generators that draw both codes from learned priors. Translation systems that deliberately retain content from an input instantiate the same analysis with that encoded-content law in place of  $p(z_c | y)$ ; term 3 is therefore sampler-specific rather than silently assumed. Our direct term-3 experiment concerns F-CS-WAE, whose stated sampler does draw  $p(z_c | y)$ .

### 3.1 COMPATIBILITY, TRANSPORT, AND COMPETENCE

The relevant object is not either marginal in isolation, but the discrepancy between  $Q_y$  and  $P_y$ . For a possibly stochastic decoder Markov kernel  $G(dx \mid z_c, z_s)$ , *transport compatibility* is  $G_{\#}Q_y = G_{\#}P_y$ . Let  $R_y = p_{\text{data}}(x \mid y)$  denote the real conditional data law. *Generation competence* asks whether  $G_{\#}P_y$  agrees with  $R_y$  under task-relevant fidelity, diversity, and condition-retention metrics. We test the latent equality first, then transport and competence separately.

**Theorem 1** (Factorized-sampling contract decomposition). *Assume the displayed regular conditional distributions exist. Define the following extended non-negative quantity:*

$$M_{\text{fact}} := \mathbb{E}_{q(y)} [\text{KL}(q(z_c, z_s \mid y) \parallel p(z_c \mid y)p(z_s))]. \quad (4)$$

Then the following identity holds in  $[0, \infty]$ :

$$\begin{aligned} M_{\text{fact}} = & \underbrace{\text{KL}(q(z_s) \parallel p(z_s))}_{(1) \text{ style-prior mismatch}} + \underbrace{I_q(z_s; y)}_{(2) \text{ condition leakage}} \\ & + \underbrace{\mathbb{E}_{q(y)} \text{KL}(q(z_c \mid y) \parallel p(z_c \mid y))}_{(3) \text{ content-prior mismatch}} + \underbrace{I_q(z_c; z_s \mid y)}_{(4) \text{ within-condition code dependence}}. \end{aligned} \quad (5)$$

All four terms are non-negative. If  $M_{\text{fact}} < \infty$ , then all four terms are finite and Eq. 5 is an ordinary real-valued equality.

The decomposition is an identity in KL, whereas our audit tests the zero-set of each term rather than estimating its numerical magnitude. The theorem allows discrete or continuous  $y$ ; our class-stratified MMD and within-class HSIC implementation targets categorical conditions with repeated samples.

**Corollary 1** (Sampler compatibility implies transport compatibility).  $M_{\text{fact}} = 0$  iff  $q(z_c, z_s \mid y) = p(z_c \mid y)p(z_s)$  for  $q(y)$ -almost every  $y$ . Moreover, for any fixed decoder Markov kernel  $G$ ,

$$M_{\text{dec}} := \mathbb{E}_{q(y)} \text{KL}(G_{\#}q(\cdot \mid y) \parallel G_{\#}[p(z_c \mid y)p(z_s)]) \leq M_{\text{fact}}. \quad (6)$$

The implication is one-way. A decoder can discard mismatched latent directions, so  $M_{\text{fact}} = 0$  is sufficient but not necessary for  $M_{\text{dec}} = 0$ . Moreover, neither latent nor decoder transport compatibility establishes agreement with real data. Competence metrics are therefore complementary to the latent decomposition and decoder interventions.

Exact equality is a falsifiable reference null, not by itself a practical severity threshold. For a pre-declared distribution-separating discrepancy  $d$  on the relevant representation space, define

$$\Delta_d := \mathbb{E}_{q(y)} d(Q_y, P_y), \quad (d, \epsilon)\text{-compatibility: } \Delta_d \leq \epsilon. \quad (7)$$

The tolerance is necessarily metric-, scale-, and task-specific; we do not claim a universal  $\epsilon$ . Equality tests establish whether the zero anchor is tenable, while raw statistics, null calibration, relative effects, and decoder/task evidence assess practical magnitude. For paired interventions we report  $R_d = 1 - \hat{\Delta}_d^{\text{int}} / \hat{\Delta}_d^{\text{ctrl}}$  when the control estimate is positive, alongside competence rather than treating rejection alone as an operational certificate.

Equation 2 constrains only term 1. Terms 2–4 concern how aggregate mass is partitioned by the condition, whether content follows its requested conditional prior, and whether the codes remain coupled within a condition. They are non-negative but not interchangeable: removing term 2 still permits the decoder to exploit term-4 dependence. Table 1 maps these clauses to held-out evidence; a direct test then checks the complete joint equality without attributing its failure to one component.

### 3.2 WHY ONE CLAUSE CANNOT CERTIFY THE CONTRACT

The following minimal counterexample isolates why exact satisfaction of term 1 leaves term 2—and therefore the joint contract—unconstrained.

**Proposition 1** (No clause-1 certificate for clause 2). *Let  $z_s \sim \mathcal{N}(0, I)$  and let  $\{A_k\}_{k=1}^K$  be a measurable partition with  $P(z_s \in A_k) = 1/K$ . Define  $y = k$  iff  $z_s \in A_k$ . Then  $q(z_s) = \mathcal{N}(0, I)$  exactly for every marginal divergence  $D$ , while  $I(z_s; y) = H(y) = \log K$ .*

Table 1: The four latent requirements and their held-out evidence. These kernel statistics are calibrated proxies for distinct equalities, not additive estimates of the four KL terms.

| Term | Failure source              | Required equality                        | Evidence used here                   |
|------|-----------------------------|------------------------------------------|--------------------------------------|
| 1    | Style-prior mismatch        | $q(z_s) = p(z_s)$                        | global MMD                           |
| 2    | Condition leakage           | $q(z_s   y) = q(z_s)$                    | probe, HSIC, $\Delta_{\text{inter}}$ |
| 3    | Content-prior mismatch      | $q(z_c   y) = p(z_c   y)$                | conditional content MMD              |
| 4    | Within-condition dependence | $q(z_c, z_s   y) = q(z_c   y)q(z_s   y)$ | within-class permutation HSIC        |

The construction is intentionally extremal, but the issue is not restricted to discontinuous partitions. Let  $z \sim \mathcal{N}(0, 1)$  and  $P(y = 1 | z) = \sigma(az)$ . Symmetry leaves the marginal of  $z$  exactly Gaussian, while  $q(z | y = 1) = 2\phi(z)\sigma(az)$  and  $q(z | y = 0) = 2\phi(z)\sigma(-az)$  are smooth, full-support, and increasingly distinguishable as  $a$  grows. Section 5.1 instantiates this construction only to establish clause-wise insufficiency; the trained-model experiments instead evaluate whether the complete audit distinguishes model regimes. Proofs are in Appendix A.

## 4 TERM-ALIGNED AUDIT AND CASE STUDY

### 4.1 CONTRACT-FIRST, TERM-ALIGNED DIAGNOSTICS

The audit follows diagnosis order: first test  $Q_y = P_y$ ; only after rejection localize the source.

The audit has two latent stages. First, independently sampled encoded and prior banks test the complete replacement without attribution:

$$T_{\text{joint}} = \sum_{k \in \mathcal{K}} \omega_k \text{MMD}^2(q(z_c, z_s | k), p(z_c | k)p(z_s)), \quad (8)$$

using a product content–style kernel and within-class pooled-label permutations. This diagnostic was specified after the original checkpoint audit and is applied to frozen checkpoints and test populations, without retraining or model selection. If it rejects, the tests in Table 1 localize the difference: global MMD for term 1; held-out probes and label-permutation HSIC for term 2; classwise content-prior MMD for term 3; and within-class permutation HSIC for term 4. Sampling-contract tests use stochastic posterior draws, while deterministic representation probes use means. Holm correction covers classwise families. These statistics test the corresponding equality; they do not estimate the numerical KL or mutual-information term. Kernel coordinates, bandwidths, weighting, and null construction are in Appendix C.

We also report classwise style-to-prior MMD,  $\text{MMD}^2(q(z_s | y = k), p(z_s))$ , as a sensitive *composite* conditional diagnostic. It can reflect term 1, term 2, or both; global MMD separates the aggregate-prior clause, while label-HSIC and probes specifically localize condition information in style. We do not identify this conditional MMD with the numerical KL value of term 2.

The fixed audit uses 2,048 test examples and a stratified 60/20/20 probe-train/validation/test split; standardization is fit on probe training data only. Probe models, kernels, permutation counts, RNG separation, and per-seed values are specified in Appendix C.

### 4.2 CONTROLLED STRESS-TEST INSTRUMENT

We use F-CS-WAE as a fully instrumented stress-test model, not as a proposed state-of-the-art generator. Its declared sampler exposes all four contract clauses; cross-family and known-factor experiments test whether the diagnostics distinguish behavior beyond this instrument. Its modified ResNet-18 encoder produces hyperspherical content  $z_c \in \mathbb{S}^{63}$  and Gaussian style  $z_s \in \mathbb{R}^{128}$  for a residual decoder. Class-specific Spherical Cauchy priors shape content; global MMD matches sampled style to  $\mathcal{N}(0, I)$ . Reconstruction, supervised content-prior matching, aggregate content matching, and a content classifier complete the phased 300-epoch objective. Appendix B specifies the sampler, architecture, coefficients, and schedule.

The historical per-class loss  $K^{-1} \sum_k \text{MMD}^2(q(z_s | k), \mathcal{N}(0, I))$  targets conditional style matching but leaves term 4 outside the intervention. Its observed instability motivates the clause-aligned intervention evaluated only after the diagnostic and decoder evidence in Section 6.

## 5 EXPERIMENTS

**Reporting rules.** The image audits cover MNIST, Fashion-MNIST, CIFAR-10, paired  $-30^\circ / +30^\circ$  Rotated-MNIST, and Shapes3D. All headline sampling-contract diagnostics use stochastic posterior samples and held-out splits.

Unbiased  $\text{MMD}^2$  estimates may be negative under the null and are never clipped. Non-rejection is interpreted at the available test power. HSIC uses plus-one permutation calibration, with domain-level decisions rather than averaged  $p$ -values. Shapes3D applies Holm correction separately within the six content tests and six style tests. Final epochs and audit populations are fixed without selecting checkpoints on leakage metrics.

### 5.1 CAN AGGREGATE MATCHING HIDE JOINT INCOMPATIBILITY?

The decomposition makes aggregate matching a clause rather than a certificate. We first isolate that logical point by instantiating the smooth construction from Proposition 1 in 128 dimensions:

$$z \sim \mathcal{N}(0, I), \quad p(y = 1 | z) = \sigma(2\kappa z_1). \quad (9)$$

The latent is sampled before the label, so  $q_\kappa(z) = \mathcal{N}(0, I)$  exactly for every  $\kappa$ : term 1 is zero by the data-generating process, not because a test fails to reject. We use 50 replications of 2,048 samples, 499 MMD and HSIC permutations, and the same stratified 60/20/20 linear-probe protocol as the checkpoint audit.

Probe accuracy rises from  $49.58 \pm 1.86\%$  at  $\kappa = 0$  to  $92.30 \pm 1.51\%$  at  $\kappa = 16$ . HSIC rejects in 100% of replications for every  $\kappa > 0$ , while its null rejection rate is 2%. Mean conditional MMD rises from  $6.16 \times 10^{-6}$  to  $4.78 \times 10^{-4}$ . These results support the intended existential claim: exact marginal equality can coexist with more than 90% recoverable label information. The complete acceptance audit and sample-size sensitivity are reported in Appendix F. This is an exact-marginal counterexample, not a trained-model result.

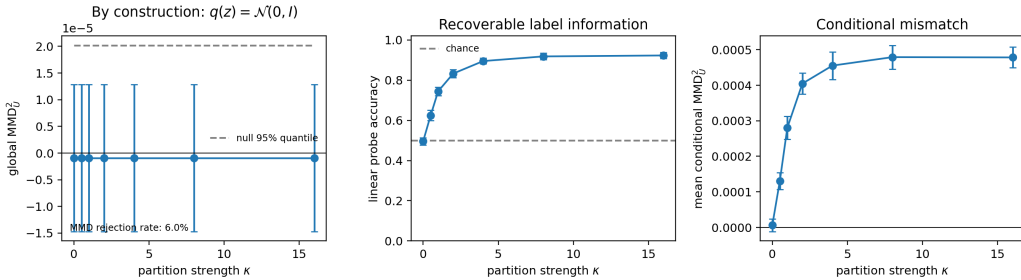


Figure 2: **Exact marginal matching does not certify condition independence.** By construction the same Gaussian latent sample is reused as  $\kappa$  changes, so global MMD remains at its null while recoverable label information and conditional mismatch increase. Points and bars show mean  $\pm$  one sample standard deviation over 50 replications.

A sweep over  $n = 512$ –4096 preserves 92.0–92.5% probe accuracy and 5–6% global-MMD rejection (Table 12). This is the expected pattern: at fixed  $\kappa$ , more observations stabilize rather than strengthen the population predictive relation, while a valid level-.05 test should remain near nominal size because the marginal null is exactly true.

### 5.2 DOES THE COMPLETE CONTRACT FAIL IN TRAINED MODELS?

Both the direct joint test and the content-prior test reject in all six seed-domain audits. The direct test localizes rejection in all 60 class-level cases, while the content-prior diagnostic rejects in 56/60.

These post-specified tests use frozen checkpoints and populations; multiplicity and independent-bank details are given in Appendix I.

Table 2: Complete-contract results on six F-CS-WAE seed-domain audits. Statistics are mean  $\pm$  sample standard deviation. Aggregate rejections remain 6/6 for each diagnostic after correction over all 12 aggregate tests. The two MMD statistics use different kernels and are not magnitude-comparable.

| Diagnostic                 | Mean statistic      | Aggregate rejects | Localized rejects |
|----------------------------|---------------------|-------------------|-------------------|
| Content-prior (term 3)     | .00807 $\pm$ .00347 | 6/6               | 56/60             |
| Sampler-compatibility test | .01943 $\pm$ .00239 | 6/6               | 60/60             |

The direct result establishes failure of the full F-CS-WAE latent replacement; the term-aligned diagnostics show that content-prior mismatch contributes alongside condition leakage and within-condition dependence.

### 5.3 DOES THE AUDIT DISTINGUISH MODEL-FAMILY REGIMES?

We next ask whether diagnostics shared across model families report leakage indiscriminately or separate practically different regimes. We train native-family, common-backbone adaptations of F-CS-WAE, DRIT (Lee et al., 2018), and DIVA (Ilse et al., 2020) for 100 epochs on the same two-domain Rotated-MNIST population. The audited style variables are F-CS-WAE  $z_s$ , DRIT’s domain-specific attribute code, and DIVA’s residual  $z_x$ . DRIT domains are audited separately because their attribute spaces are domain-specific.

Table 3: Cross-model audit on a common Rotated-MNIST backbone, mean  $\pm$  sample standard deviation over three seeds. Accuracies and retention are percentages. Raw MMD values describe the fixed audit geometry, not a model-quality ranking; an unbiased estimate may be negative under its null.

| Family   | Global MMD               | Style-prior MMD       | Style $\rightarrow$ label | Content acc.     | Rec. L1             | Prior retain     |
|----------|--------------------------|-----------------------|---------------------------|------------------|---------------------|------------------|
| F-CS-WAE | .003326 $\pm$ .000140    | .032262 $\pm$ .001197 | 91.92 $\pm$ 2.43          | 99.42 $\pm$ .14  | .01510 $\pm$ .00036 | 12.37 $\pm$ 1.19 |
| DRIT     | .255261 $\pm$ .008545    | .272442 $\pm$ .001938 | 34.42 $\pm$ 12.62         | 88.25 $\pm$ 1.52 | .04284 $\pm$ .00211 | 9.87 $\pm$ .12   |
| DIVA     | -.0000179 $\pm$ .0000274 | .000176 $\pm$ .000129 | 10.67 $\pm$ 1.04          | 96.67 $\pm$ .76  | .03449 $\pm$ .00009 | 89.45 $\pm$ .83  |

The cross-family subset separates three qualitatively different outcomes. F-CS-WAE has a larger conditional than global discrepancy and high, seed-stable label recovery. DRIT’s global discrepancy is already large, so its label-bearing attribute code is a broad prior-sampler mismatch rather than a matched-marginal paradox. DIVA has near-null global discrepancy, chance-level probes, and competent prior-style retention. HSIC rejects in all six seed-domain tests for F-CS-WAE and DRIT ( $p = 1/201$ ). Two of six raw DIVA tests fall below .05, but none survives Holm correction within the DIVA family and the probe pattern does not repeat. DIVA therefore provides a near-null regime at the available power.

All three adaptations pass the pre-specified content-representation gate, making a trivial collapse of the content code unlikely to explain the contrast. These common-backbone adaptations test audit discrimination under shared conditions; their full probe suite, frozen competence gates, and relation to each family’s native generation law are documented in Appendix H.

### 5.4 DOES THE MISMATCH SURVIVE DECODING?

**Decoded semantics.** The data-processing inequality leaves open whether a latent mismatch affects outputs. We therefore pair the fixed MNIST, Fashion-MNIST, and CIFAR-10 audit with prior-sampling and donor-swap interventions. Conditional MMD is .0160 on MNIST and .010 on Fashion-MNIST, and all seedwise HSIC tests attain  $p = 1/201$ . Replacing global Gaussian style by post-hoc class styles raises MNIST Gen-ACC from 10.47% to 100%; deterministic mean swaps similarly make decoded identity follow the style donor in 99.28% of MNIST and 63.92% of

CIFAR-10 cases. An archived per-class intervention lowers donor following to 23.37% and 35.92%, respectively, while within-class HSIC continues to reject, consistent with suppressing term 2 but not term 4.

The variance head of F-CS-WAE often reaches its numerical floor, but a three-seed variance-floor intervention does not fully explain the effect: increasing posterior noise improves independent generation while substantial MNIST label recovery remains. The complete intervention, CIFAR-10 contrast, legacy classwise-style result, qualitative grids, and per-level statistics are reported in Appendices G and D. Section 6 next tests whether directly optimizing the diagnosed clauses changes both latent statistics and decoder attribution.

### 5.5 DOES THE AUDIT RECOVER KNOWN GENERATIVE FACTORS?

On an IID Shapes3D split, every model passes reconstruction/content gates, but F-CS-WAE style predicts shape at 99.0% and content predicts nuisance factors at 87.0% on average. The Holm-corrected pattern repeats over three seeds: F-CS-WAE rejects all six factor tests in both views, while a learned content–style VAE rejects only the expected complementary factors and preserves shape at 99.7%/99.8% under prior/swap decoding, versus 89.0%/88.0% for F-CS-WAE. Appendix J reports the second near-null control, oracle scope, and per-factor results.

**Released-checkpoint boundary.** Appendix K separately audits SCFlow’s released synthetic CLIP-space checkpoint (Ma et al., 2025). Its endpoint law differs from the four-term encoder–prior setting, so we use it only to illustrate that improved forward retrieval can coexist with endpoint/inverse-transport failure, not as a natural-image validation of Theorem 1.

## 6 CLAUSE-ALIGNED INTERVENTION

We treat factorized-contract training (FACT) as a mechanism probe, not a standalone optimizer or end-to-end generation method.

**Intervention.** After warm-up, FACT replaces legacy classwise penalties with positive null-excess constraints for conditional style MMD, conditional content MMD, and within-class sampled-code HSIC, calibrated against matched prior–prior banks or within-class permutations. Sampled-style label HSIC, an adversarial classifier, and posterior-mean HSIC target the remaining nonlinear information. Content is stop-gradient for dependence terms, and direct joint MMD remains a held-out outcome. Appendix L gives the exact objective, optimization, calibration, and development history. Raw/null calibration and component construction use seed 0; the setting and seven gates are then fixed before seeds 1–2, with training seeds separated from audit RNGs.

**Fully traceable matched result.** We report every retained control/FACT pair for training seeds 0–2; seed 0 is the development pair and seeds 1–2 were run after the setting was fixed. Joint MMD falls from  $.002368 \pm .000160$  to  $.001449 \pm .000166$ , a  $38.89 \pm 3.65\%$  paired reduction; classwise style–prior MMD falls from  $.002567 \pm .000153$  to  $.001902 \pm .000148$ . Nevertheless, all three controls and all three FACT models reject joint equality ( $p = .005$  in every aggregate test). Style-probe accuracy falls from 76.91% to 11.14%. Decoder attribution moves in the intended direction in every pair: content-donor following rises by  $25.41 \pm 8.05$  points and style-donor following falls by  $29.09 \pm 5.32$  points. These gains carry a cost: global Gen-ACC falls by  $9.93 \pm 2.84$  points and reconstruction MAE worsens by  $6.52 \pm 5.57\%$ .

Table 4: Matched MNIST FACT–control effects over seeds 0–2 (mean  $\pm$  sample SD; all seeds retained).

| Outcome                          | Mean $\pm$ SD     | Direction | Seeds agreeing |
|----------------------------------|-------------------|-----------|----------------|
| Joint MMD reduction (%)          | $38.89 \pm 3.65$  | lower     | 3/3            |
| Content-donor following (points) | $+25.41 \pm 8.05$ | higher    | 3/3            |
| Style-donor following (points)   | $-29.09 \pm 5.32$ | lower     | 3/3            |
| Global Gen-ACC (points)          | $-9.93 \pm 2.84$  | worse     | 3/3            |
| Reconstruction MAE (relative %)  | $+6.52 \pm 5.57$  | worse     | 3/3            |

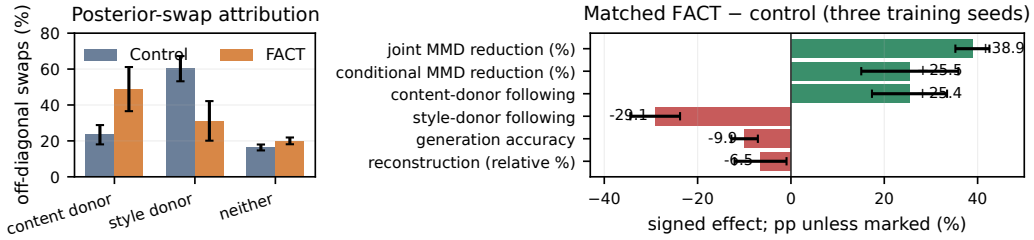


Figure 3: **Selected-diagnostic gains do not imply competence gains.** Matched posterior-swap attribution and paired effects (mean  $\pm$  sample SD); green favors the intended target and red marks competence costs.

**Metric robustness and scope.** MMD and a post-specified energy check agree on 98.99% of equality decisions, and distance correlation falls in every pair. Yet joint energy improves in only two pairs and worsens sharply in seed 0. Thus the selected-MMD reduction and swap change neither establish metric-independent progress nor make the exact joint null tenable; Appendix L gives all outcomes.

## 7 DISCUSSION AND LIMITATIONS

**How to use the audit.** The declared sampler determines the reference law. Test  $Q_y = P_y$  jointly, localize rejection with the four clauses, then measure decoder/task effects. The clauses point to the aggregate prior, style leakage, content sampler, and residual dependence. Decoder-insensitive mismatch may need no change; useful label-bearing style may instead call for changing the sampler.

**Equality versus practical compatibility.** Equality rejection localizes failure; it is not a quality verdict. Deployment claims should pre-declare  $(d, \epsilon)$  and report  $\Delta_d$  with transport and competence. SCFlow’s retrieval gain despite endpoint mismatch and FACT’s opposing MMD and Gen-ACC changes rule out a universal threshold; cross-architecture tolerance calibration remains open.

**Intervention scope.** The three pairs support a stable direction for the chosen MNIST MMD and swap metrics, not a general optimizer. Kernel-free magnitude and competence do not improve uniformly, so diagnostics require explicit transport and task constraints.

**Limitations.** Shapes3D is IID, DRIT/DIVA are adapted, and SCFlow is a CLIP-space boundary rather than a four-term natural-image audit. Decoded natural-image and continuous-condition evidence remains open. FACT is sequentially built on MNIST from only three traceable pairs and needs factorial confirmation.

## 8 CONCLUSION

Factorized generation replaces the paired reconstruction law by an independent latent law, while aggregate matching verifies only one of four requirements. Mismatch can alter decoding, yet optimizing selected diagnostics can harm generation; compatibility, transport, and competence must be evaluated separately.

## REPRODUCIBILITY STATEMENT

Section 4 specifies the latent views, preprocessing, tests, permutation calibration, and split discipline. Sections 5.1– 6 state setting-specific populations, seeds, analysis status, and fixed reporting rules. The appendix contains complete proofs, architectures, per-seed results, multiplicity rules, competence gates, and limitations for each external audit. The anonymous supplementary package ties unrounded headline rows, figures, split/protocol metadata, and critical source files to a content manifest; checkpoint hashes are retained without claiming that the large checkpoint files are packaged. Appendix M collects the evidence roots and boundaries.

## ETHICS STATEMENT

This work uses standard public computer-vision benchmarks and existing model checkpoints, collects no new data, and involves no human subjects or sensitive personal information. Its purpose is to identify and measure a failure mode in factorized generative models. Generated examples are low-resolution benchmark images; nevertheless, the broader recommendation—auditing whether a decoder uses unintended latent information—may also help identify representation leakage in higher-stakes applications.

## AI USE STATEMENT

Generative AI tools assisted with feedback on methodology and experiment design; implementation, debugging, and review of loss, audit, and orchestration code; interpretation of results; identification and summarization of related literature; and drafting, restructuring, and editing the manuscript. They were not used to generate datasets or fabricate experimental measurements. The authors manually inspected the cited primary literature, verified every mathematical claim, executed tests and experiments, checked source and artifact hashes and every reported numerical value, and reviewed all AI-assisted code and text. The authors take responsibility for the final content, claims, code, and artifacts.

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## A PROOFS

### A.1 PROOF OF PROPOSITION 1

The label is defined after drawing  $z_s$  and therefore does not alter the marginal law of  $z_s$ . Hence  $q(z_s) = \mathcal{N}(0, I)$  and any divergence between these two identical measures is zero. Because  $y$  is a deterministic function of  $z_s$ ,  $H(y | z_s) = 0$ . The equal-probability partition gives  $H(y) = \log K$ , and consequently  $I(z_s; y) = H(y) - H(y | z_s) = \log K$ .

## A.2 A SMOOTH, FULL-SUPPORT CONSTRUCTION

Let  $z \sim \mathcal{N}(0, 1)$  and, conditional on  $z$ , draw a binary label with

$$P(y = 1 | z) = \sigma(az), \quad P(y = 0 | z) = \sigma(-az), \quad (10)$$

where  $a > 0$  and  $\sigma(t) = (1 + e^{-t})^{-1}$ . Since the standard normal density  $\phi$  is symmetric and  $\sigma(t) + \sigma(-t) = 1$ ,

$$P(y = 1) = \int_{\mathbb{R}} \phi(z) \sigma(az) dz \quad (11)$$

$$= \frac{1}{2} \int_{\mathbb{R}} \phi(z) [\sigma(az) + \sigma(-az)] dz = \frac{1}{2}. \quad (12)$$

The generative construction starts with the standard normal and changes only the conditional label law, so the marginal of  $z$  remains exactly standard normal. Bayes' rule gives

$$q(z | y = 1) = 2\phi(z)\sigma(az), \quad q(z | y = 0) = 2\phi(z)\sigma(-az). \quad (13)$$

Both densities are smooth and strictly positive on  $\mathbb{R}$ . When  $a \rightarrow \infty$ , the conditional label approaches the indicator of the sign of  $z$  outside a set of Gaussian measure zero. Thus  $H(y | z) \rightarrow 0$  and  $I(z; y) \rightarrow \log 2$  while the marginal remains unchanged.

## A.3 PROOF OF THEOREM 1

For a fixed  $y$ , apply the relative-entropy chain rule to the encoded joint and the product reference measure:

$$\text{KL}(q(z_c, z_s | y) \| p(z_c | y)p(z_s)) \quad (14)$$

$$= \text{KL}(q(z_c, z_s | y) \| q(z_c | y)q(z_s | y)) \\ + \text{KL}(q(z_c | y) \| p(z_c | y)) + \text{KL}(q(z_s | y) \| p(z_s)). \quad (15)$$

The first term is  $I_q(z_c; z_s | y)$  at that label. Averaging over  $q(y)$  gives terms 3 and 4 of Eq. 5. For the remaining style term, insert the aggregate posterior  $q(z_s)$ :

$$\mathbb{E}_{q(y)} \text{KL}(q(z_s | y) \| p(z_s)) = \mathbb{E}_{q(y)} \text{KL}(q(z_s | y) \| q(z_s)) \\ + \text{KL}(q(z_s) \| p(z_s)) \quad (16)$$

$$= I_q(z_s; y) + \text{KL}(q(z_s) \| p(z_s)). \quad (17)$$

These are identities of non-negative extended-real quantities and require no subtraction of infinities. Applying the same chain rule to the joint law  $q(y, z_s)$  and reference  $q(y)p(z_s)$  yields the style identity above, so the full decomposition holds in  $[0, \infty]$ . If  $M_{\text{fact}} < \infty$ , the non-negative summands on the right are bounded by their finite sum; hence all four are finite and the identity is ordinary real-valued equality. Non-negativity follows from non-negativity of relative entropy and mutual information.

## A.4 PROOF OF COROLLARY 1

An expectation of a non-negative KL is zero iff its integrand is zero for  $q(y)$ -almost every  $y$ , which holds iff the two latent conditional measures are equal. The pushforward inequality follows by applying the data-processing inequality to the decoder Markov kernel for each  $y$ , then averaging over  $q(y)$ . This covers deterministic and stochastic decoders.

## B ARCHITECTURE AND TRAINING

### B.1 ARCHITECTURE

The shared encoder is a ResNet-18 adapted to small images: its first  $7 \times 7$ , stride-2 convolution is replaced by a  $3 \times 3$ , stride-1 convolution and max pooling is removed. The trunk output is flattened and mapped to 256 dimensions with SiLU. Separate heads produce an unnormalized semantic direction, a scalar semantic concentration, a style mean, and a style log variance. We use

$$\mu_c = \widetilde{\mu}_c / \|\widetilde{\mu}_c\|_2, \quad \rho_c = \sigma(r_c)(1 - \varepsilon), \quad (18)$$

$$z_s = \mu_s + \exp(\tfrac{1}{2} \log \sigma_s^2) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (19)$$

Here  $\varepsilon = 10^{-8}$  only enforces  $\rho_c < 1$ . With  $e \sim \text{Unif}(\mathbb{S}^{d-1})$ , the implemented Spherical Cauchy sample is the normalized Möbius transform

$$z = \text{norm}\left(\frac{(1 - \rho^2)e + 2\rho^2\mu + 2\rho(e^\top \mu)\mu}{1 + 2\rho e^\top \mu + \rho^2 + \varepsilon}\right), \quad (20)$$

following the efficient Spherical Cauchy construction of Sablica & Hornik (2025). The decoder concatenates  $[z_c; z_s]$ , projects it to a  $512 \times 4 \times 4$  tensor, and applies three residual bilinear-upsample blocks with channel widths 256, 128, 64. Its sigmoid output is interpolated to the native dataset resolution when necessary. MNIST and Fashion-MNIST therefore remain  $28 \times 28$  and single-channel; CIFAR-10 remains  $32 \times 32 \times 3$ .

For class  $k$ , the semantic prior is a Spherical Cauchy distribution centered at an EMA direction  $m_k$ , with concentration  $\rho_p = 0.7$ . The center is updated once per epoch from encoded class-mean directions with momentum 0.95. The style prior is  $\mathcal{N}(0, I)$ .

## B.2 OBJECTIVE

The training objective is

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} \\ & + \delta(t)\mathcal{L}_{\text{style-cl}} + \eta(t)\mathcal{L}_{\text{cls}}. \end{aligned} \quad (21)$$

$\mathcal{L}_{\text{rec}}$  is L1 plus 0.1 LPIPS (Zhang et al., 2018).  $\mathcal{L}_{\text{class}}$  matches each semantic class conditional to its spherical prior,  $\mathcal{L}_{\text{agg}}$  matches aggregate semantic codes to the class-prior mixture,  $\mathcal{L}_{\text{style}}$  matches sampled style codes to a standard Gaussian, and  $\mathcal{L}_{\text{cls}}$  is cross entropy from  $\mu_c$ . The baseline checkpoints audited in the main paper explicitly set  $\delta = 0$ ; the repository default for new remedied runs is  $\delta = 1$ , so the run manifest or command-line override is required to distinguish them.

Table 5: Case-study training hyperparameters.

| Parameter                           | Value                         |
|-------------------------------------|-------------------------------|
| Optimizer                           | Adam, learning rate $10^{-3}$ |
| Schedule                            | StepLR, step 100, factor 0.5  |
| Batch size / epochs                 | 128 / 300                     |
| Gradient clipping                   | global norm 1.0               |
| Semantic / style dimension          | 64 / 128                      |
| Semantic-prior concentration        | 0.7                           |
| EMA momentum                        | 0.95                          |
| Final class / aggregate MMD weights | 2.0 / 5.0                     |
| Final global style MMD weight       | 1.0                           |
| Final auxiliary-classifier weight   | 0.3                           |

Training has four phases: reconstruction only for epochs 0–49; global style MMD and the auxiliary classifier ramp during epochs 50–99; semantic class and aggregate MMD ramp to half strength during epochs 100–199; all terms reach full strength during epochs 200–299. For remedied runs, per-class style MMD follows the global-style ramp. No image augmentation is used.

## C CANONICAL AUDIT PROTOCOL

### C.1 DATA SPLIT AND LATENT VIEWS

For each checkpoint we select 2,048 examples from the standard test split with evaluation seed 0. Probe indices are generated by a deterministic stratified 60/20/20 split, yielding 1,228 training, 410 validation, and 410 test examples. Standardization parameters are estimated only on the 1,228 probe-training examples. The split indices are stored by SHA-256 hash in each result manifest.

The encoder is evaluated once to obtain  $(\mu_c, \mu_s, \log \sigma_s^2)$  and a seeded stochastic draw  $z_s$ . We use:

- $\mu_s$  for representation probes and deterministic mean swaps;

- $z_s$  for global MMD, conditional style statistics, and the sampling-contract probe; and
- both  $(\mu_c, \mu_s)$  and repeated posterior draws  $(z_c, z_s)$  for the conditional-dependence diagnostic.

Agreement between these views is reported as an empirical result rather than assumed.

## C.2 PROBE SUITE

The suite contains multinomial logistic regression, a fixed two-layer MLP, RBF-SVM, and  $k$ -NN. Validation data are used for fixed early-stopping or model-selection rules; every headline accuracy is evaluated once on the untouched test partition. We report accuracy and macro-F1. The main paper uses logistic regression as the simplest standardized headline and includes RBF-SVM to detect nonlinear separation.

## C.3 MMD AND HSIC

Global MMD is the unbiased U-statistic with the averaged Euclidean RBF kernel

$$k(u, v) = \frac{1}{7} \sum_{\sigma \in \{0.5, 1, 2, 5, 10, 20, 50\}} \exp\left(-\frac{\|u - v\|_2^2}{2\sigma^2}\right). \quad (22)$$

The Gaussian reference has the same sample count as the encoded set and uses an RNG stream independent of posterior sampling. A balanced pooled-label permutation test calibrates the finite-sample null with 500 permutations. The resulting null 95th percentiles are  $1.88\text{--}2.45 \times 10^{-5}$ , whereas observed global MMD is  $2.48\text{--}5.79 \times 10^{-4}$ ; exact equality is rejected for every checkpoint ( $p = 1/501$ ). We therefore use “small” only as a relative numerical description, not as acceptance of the equality null.  $\Delta_{\text{inter}}$  is not an MMD; it is the average Euclidean distance between all pairs of class-conditional style means.

Label HSIC centers a multiscale style Gram matrix and a label Gram matrix and uses 1,024 examples for bounded memory. We calibrate it by 200 label permutations. With the plus-one correction,

$$p = \frac{1 + \#\{T_b \geq T_{\text{obs}}\}}{201}, \quad (23)$$

so  $p = 0.004975$  is reported rather than  $p = 0$ . For the within-class permutation statistic, with represented classes  $\mathcal{K}$  and empirical weights  $\omega_k$ , we use

$$T_{\text{WC}} = \sum_{k \in \mathcal{K}} \omega_k \text{HSIC}(z_c^{(k)}, z_s^{(k)}). \quad (24)$$

Content vectors are normalized to the unit sphere and styles are standardized within class. We use the chordal multiscale kernel with bandwidths  $\{.1, .3, .5, 1, 2\}$  for content and the Euclidean ladder above for style. Each classwise Gram matrix is centered and classwise HSIC is divided by  $(n_k - 1)^2$ . The reported statistic weights classes by  $\omega_k = q(y = k)$ , estimated as  $n_k / \sum_j n_j$ , and every null replicate independently permutes the style Gram matrix within each class before recomputing the same weighted sum. Because every audit population used here is balanced, this equals the unweighted macro average numerically.

The historical JointMMD proxy paired each observation with a permutation of the same style bank and is therefore not used as a headline independent two-sample test. The split-bank diagnostic in Appendix I resolves this issue: posterior and prior banks are independently sampled, the content-prior MMD uses the same chordal ladder, and the direct-joint MMD uses the product of content and style kernels. Both aggregate their balanced class statistics before permutation calibration; per-class decisions use Holm correction.

## D FULL STAGE-0 RESULTS

For term 4, ten stochastic posterior draws per checkpoint all attain the minimum calibrated  $p = 1/201$ . The sample-view statistic is  $0.002682 \pm 0.000005$  across draws on MNIST,  $0.002459$  on

Table 6: Full held-out probe suite from the original audit. Values are test accuracy in percent. MNIST/Fashion-MNIST rows are seed-0 details; CIFAR-10 is mean  $\pm$  sample standard deviation over model seeds 0, 1, 2.

| Dataset  | View    | Logistic         | MLP              | RBF-SVM          | k-NN             |
|----------|---------|------------------|------------------|------------------|------------------|
| MNIST    | $\mu_s$ | 99.76            | 99.51            | 99.76            | 97.80            |
|          | $z_s$   | 99.76            | 99.51            | 99.76            | 97.80            |
| Fashion  | $\mu_s$ | 88.29            | 90.73            | 93.17            | 85.12            |
|          | $z_s$   | 88.29            | 90.73            | 93.17            | 85.61            |
| CIFAR-10 | $\mu_s$ | $68.46 \pm 6.29$ | $70.49 \pm 3.83$ | $70.24 \pm 3.23$ | $42.52 \pm 6.93$ |
|          | $z_s$   | $68.62 \pm 6.19$ | $70.57 \pm 3.50$ | $70.16 \pm 3.36$ | $42.20 \pm 7.11$ |

Table 7: Completed three-seed headline replication for the original MNIST and Fashion-MNIST checkpoints. Values are rounded aggregate results; every seedwise HSIC test attains  $p = 1/201$ .

| Dataset       | Global MMD | Logit $z_s$ (%) | Cond. MMD | HSIC $p$ |
|---------------|------------|-----------------|-----------|----------|
| MNIST         | .00058     | 99.6            | .0160     | .00498   |
| Fashion-MNIST | .00034     | 88.0            | .010      | .00498   |

Fashion-MNIST, and has per-model means 0.002250, 0.002292, and 0.002338 on CIFAR-10. The seed-0  $\delta = 1$  checkpoints also reject on all ten draws, with means 0.002507 (MNIST) and 0.002283 (CIFAR-10). These calibrated kernel tests are evidence of stochastic within-class dependence, not numerical estimates of  $I(z_c; z_s | y)$ .

#### D.1 STRUCTURE OF THE CONDITIONAL MISMATCH

The class averages reported for the original checkpoints compress a structured ten-class discrepancy. Figure 4 compares every pair of class-conditional style distributions. Broad off-diagonal mass is visible on both datasets, rather than separation being confined to a few class pairs. These matrices use the same stochastic-view diagnostic as the per-class prior comparison; their scale should not be compared across datasets as if it were a normalized dependence measure.

Figure 5 gives a complementary one-dimensional view. The projection direction is computed from the class means rather than chosen after inspecting individual examples. Several MNIST classes separate along this direction, but the overlap also illustrates why a single projection is not a substitute for the held-out multivariate probes.

Table 8: Per-seed CIFAR-10 results for the two principal probes.

| Seed | $\Delta_{\text{inter}}(z_s)$ | Logit $\mu_s$ | Logit $z_s$ | RBF $\mu_s$ | RBF $z_s$ |
|------|------------------------------|---------------|-------------|-------------|-----------|
| 0    | 3.801                        | 61.46         | 61.71       | 66.59       | 66.34     |
| 1    | 4.264                        | 73.66         | 73.66       | 72.68       | 72.68     |
| 2    | 3.797                        | 70.24         | 70.49       | 71.46       | 71.46     |

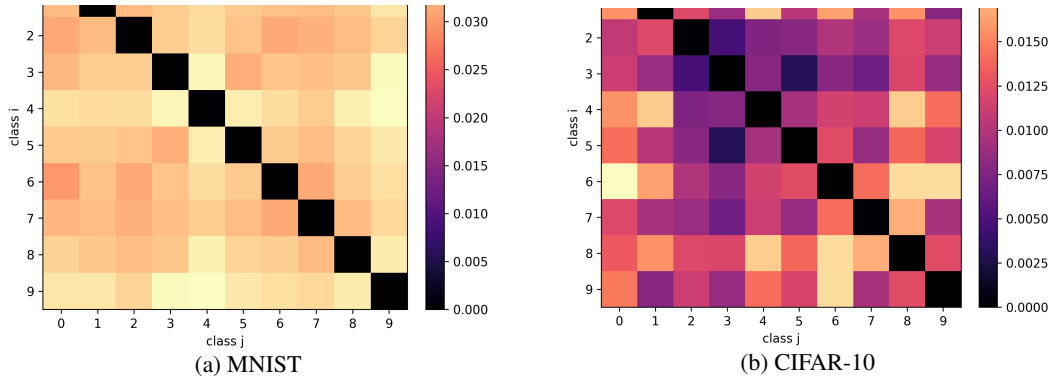


Figure 4: Pairwise  $\text{MMD}^2(q(z_s | y = i), q(z_s | y = j))$  for the baseline seed-0 checkpoints. The diagonal is zero by construction; off-diagonal structure shows that conditional style distributions differ throughout the label set.

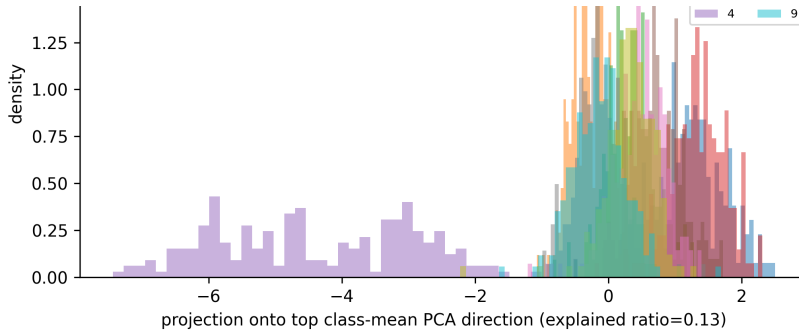


Figure 5: MNIST stochastic styles projected onto the leading direction of the class-mean matrix at  $\delta = 0$ . The direction explains 13.1% of the class-mean variance. The plot is descriptive; the held-out probes and HSIC tests provide the quantitative evidence.

## D.2 POSTERIOR-VARIANCE INSPECTION

The fractions of raw style log-variance coordinates at or below the clamp floor are 99.9966% (MNIST), 99.9989% (Fashion-MNIST), and 100% for all three CIFAR-10 checkpoints. Because this statistic was added during the post-hoc checkpoint audit, it is reported in the human-readable audit report; future manifests should also store it as a first-class numeric field.

## D.3 COMPLETE CONDITIONAL-REGULARIZATION SWEEP

Table 9 expands the four points shown in the main paper to all six archived  $\delta$  settings. We retain ACC,  $\Delta_{\text{inter}}$ , and FID because they are directly tied to the sweep artifacts, but omit the historical probe column until it is regenerated with the canonical held-out split. Each point is a separate single-seed training run.

Table 9: Complete single-seed per-class-style-MMD sweep. Lower FID and  $\Delta_{\text{inter}}$  are better; no uncertainty or significance claim is made.

| $\delta$ | MNIST |                         |      | CIFAR-10 |                         |      |
|----------|-------|-------------------------|------|----------|-------------------------|------|
|          | ACC   | $\Delta_{\text{inter}}$ | FID  | ACC      | $\Delta_{\text{inter}}$ | FID  |
| 0        | .9915 | 6.21                    | 74.0 | .8129    | 3.74                    | 80.5 |
| .03      | .9944 | 3.27                    | 62.8 | .8111    | 2.48                    | 79.1 |
| .1       | .9898 | 2.12                    | 52.6 | .8100    | 2.03                    | 77.5 |
| .3       | .9936 | 1.55                    | 59.4 | .7148    | 1.59                    | 72.0 |
| 1        | .8592 | 1.22                    | 43.4 | .8000    | 1.05                    | 74.4 |
| 3        | .9929 | 1.18                    | 41.5 | .7994    | .95                     | 76.1 |

The reduction in  $\Delta_{\text{inter}}$  is broadly monotone, but the operating-point quality is not: MNIST ACC falls to .8592 specifically at  $\delta = 1$  while the neighboring .1 and .3 runs remain at .9898 and .9936, and the  $\delta = 3$  run returns to .9929. The completed three-seed ACC ranges are .985–.994 at  $\delta = 0$ , .983–.993 at .1, .985–.995 at .3, .850–.8656 at 1, and .985–.995 at 3. The degradation at  $\delta = 1$  is therefore a replicated operating-regime effect rather than one bad seed. Because the decoder swap intervention itself is available only for seed 0 at  $\delta = 1$ , its large reduction in style-donor following remains a directional mechanism check rather than evidence for a reliable remediation setting.

## E GENERATION AND DECODER DIAGNOSTICS

### E.1 EXTERNAL GENERATION EVALUATION

The independent MNIST evaluator is a four-convolution CNN trained for 20 epochs on real images only. Its standard test accuracy is 99.34%. The CIFAR-10 evaluator is a WideResNet-28-10 trained for 200 epochs with standard random-crop and horizontal-flip augmentation on real training images only; its test accuracy is 96.16%. Neither evaluator contributes to the generative model’s loss.

Table 10: Per-checkpoint external Gen-ACC. Each entry uses 1,000 images per requested class and a class-conditional style bank fitted on 12,000 training examples.

| Dataset  | Model seed | Global Gaussian | Class diagonal $\tau = .25$ |
|----------|------------|-----------------|-----------------------------|
| MNIST    | 0          | 0.1047          | 1.0000                      |
| CIFAR-10 | 0          | 0.1262          | 0.4016                      |
| CIFAR-10 | 1          | 0.1320          | 0.5386                      |
| CIFAR-10 | 2          | 0.1268          | 0.4747                      |

The class-diagonal distribution is fit from encoded training styles and therefore deliberately conditions style on the requested label. Its gain tests whether the decoder is sensitive to the conditional mismatch. It is not an unconditional generative model and does not alter the encoder.

Figure 6 displays the qualitative counterpart of Table 10. Conditioning style makes the requested category more visually consistent in some rows, but shrinking the diagonal Gaussian to  $\tau = .25$



Figure 6: Archived CIFAR-10 samples from the seed-0 checkpoint. For four requested classes, odd rows use the global Gaussian style prior and even rows use the post-hoc class-diagonal prior at  $\tau = .25$ . The grid is qualitative; external Gen-ACC is reported in Table 10.

also reduces variation and can produce blurry or prototype-like outputs. This is why Gen-ACC is not interpreted as a stand-alone measure of generative quality; diversity-sensitive evaluation on the same generated pools remains necessary.

## E.2 INDEPENDENT-EVALUATOR DETERMINISTIC MEAN SWAPS

For each ordered pair of distinct classes  $(a, b)$ , the diagnostic decodes  $\text{Dec}(\mu_c^{(a)}, \mu_s^{(b)})$  for 500 donor pairs and evaluates pixels with the independent real-image classifier. On baseline checkpoints, identity follows the style donor on 99.28% of MNIST and 63.92% of CIFAR-10 off-diagonal swaps, while following the semantic donor on 0.06% and 4.72%. At  $\delta = 1$ , style-following falls to 23.37%

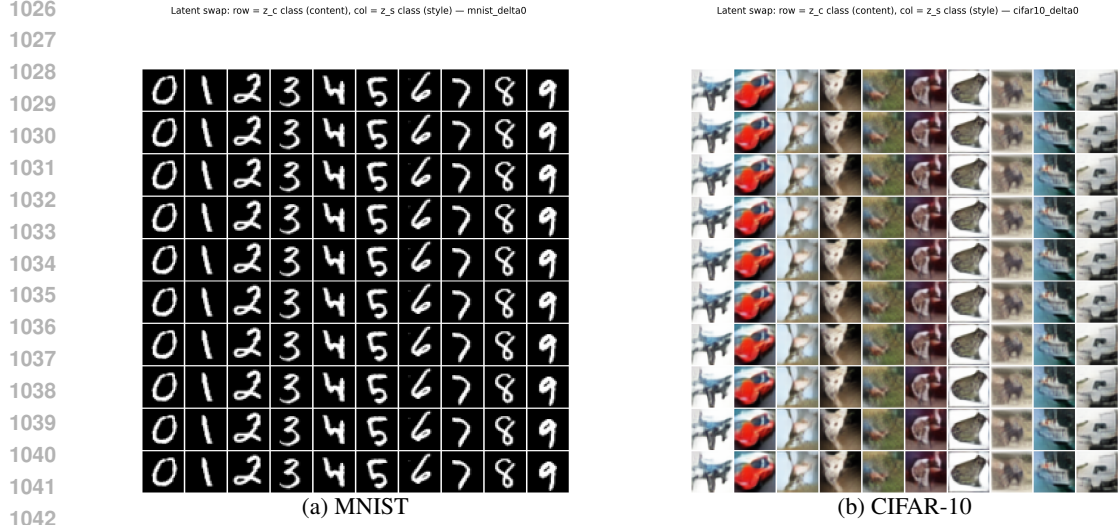


Figure 7: Deterministic mean-swap grids at  $\delta = 0$ . Rows donate semantic means and columns donate style means. Column-controlled identity is visually strong on MNIST and weaker but still present for several CIFAR-10 classes. Quantitative rates in the text use independent pixel classifiers; the grids are qualitative.

and 35.92%, and semantic- following rises to 51.00% and 18.74%, respectively. Full rate matrices, the other-class outcome, and pair-bootstrap intervals are stored in the external-swap artifacts.

## F EXACT-MARGINAL EXPERIMENT

### F.1 CONSTRUCTION AND PROTOCOL

For every replication we draw  $n = 2,048$  vectors  $z \sim \mathcal{N}(0, I_{128})$  once, then reuse that latent sample while drawing binary labels from logits  $[-\kappa z_1, \kappa z_1]$  for  $\kappa \in \{0, .5, 1, 2, 4, 8, 16\}$ . Thus the population marginal and the finite latent sample are invariant to  $\kappa$ ; only the label mechanism changes. We run 50 replications from base seed 2027, 499 global-MMD permutations, 499 HSIC permutations, and a stratified 60/20/20 linear-probe split. The independent Gaussian reference and the labels use distinct random streams. Plus-one calibration gives minimum  $p = 1/500$ .

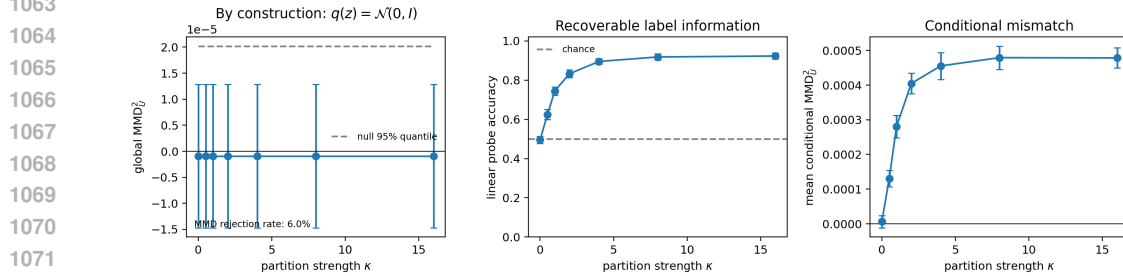


Figure 8: Exact marginal matching does not certify label independence. The same Gaussian sample is relabeled as  $\kappa$  changes, so global MMD remains invariant while probe accuracy and conditional discrepancy increase. Points and bars are mean  $\pm$  one sample standard deviation over 50 replications.

The unbiased global MMD can be negative under the null. Three of 50 global tests reject, inside the frozen central 95% prediction interval  $[0, 6]$  for Binomial(50, .05). HSIC has Spearman trend 1.0 and conditional MMD trend .964. Seven of eight frozen checks pass. The failed check required

Table 11: Complete synthetic result. Probe values are mean  $\pm$  sample standard deviation over 50 replications; rejection columns are empirical rates. The global statistic is repeated across  $\kappa$  within each replication by design.

| $\kappa$ | Probe (%)        | Global MMD             | MMD rej. | HSIC    | HSIC rej. | Cond. MMD |
|----------|------------------|------------------------|----------|---------|-----------|-----------|
| 0        | 49.58 $\pm$ 1.86 | $-9.47 \times 10^{-7}$ | .06      | .000176 | .02       | .000006   |
| .5       | 62.48 $\pm$ 2.58 | $-9.47 \times 10^{-7}$ | .06      | .000239 | 1.00      | .000130   |
| 1        | 74.49 $\pm$ 2.09 | $-9.47 \times 10^{-7}$ | .06      | .000314 | 1.00      | .000280   |
| 2        | 83.28 $\pm$ 2.07 | $-9.47 \times 10^{-7}$ | .06      | .000377 | 1.00      | .000405   |
| 4        | 89.51 $\pm$ 1.45 | $-9.47 \times 10^{-7}$ | .06      | .000404 | 1.00      | .000455   |
| 8        | 91.82 $\pm$ 1.63 | $-9.47 \times 10^{-7}$ | .06      | .000413 | 1.00      | .000479   |
| 16       | 92.30 $\pm$ 1.51 | $-9.47 \times 10^{-7}$ | .06      | .000415 | 1.00      | .000478   |

mean probe accuracy strictly above 90% at each of  $\{4, 8, 16\}$ ;  $\kappa = 4$  reaches 89.51%. No threshold, level, or inequality was changed after observing the result.

Table 12: Supplementary sample-size sensitivity at  $\kappa = 16$ . Values are rounded aggregate results from the completed follow-up runs.

| $n$   | Probe accuracy (%) | Global-MMD rejection (%) |
|-------|--------------------|--------------------------|
| 512   | 92.0               | 6.0                      |
| 1,024 | 92.2               | 5.0                      |
| 2,048 | 92.3               | 6.0                      |
| 4,096 | 92.5               | 5.0                      |

The probe effect is essentially stable across an eightfold range in sample size, whereas the rejection rate remains near the nominal 5% level. The follow-up therefore supports a variance-reduction interpretation rather than an effect-size increase with  $n$ .

## G CONTROLLED POSTERIOR-ENTROPY INTERVENTION

The intervention adds a fixed floor in variance space and uses the resulting  $\sigma_{\text{eff}}$  in both training and audit. A floor of zero reproduces the historical sampler. Every reported setting is trained for 300 epochs with MNIST and CIFAR-10 model seeds  $\{0, 1, 2\}$ ; the original seed-0 sweep below is retained at full precision and the completed summaries follow it. The CIFAR-10 confirmation levels  $\{0, .15, .35\}$  were fixed after both .15 and .35 passed the pre-specified MNIST promotion rule. That rule required an entropy increase of at least .75 nats/dimension, probe accuracy at least .20, HSIC rejection, conditional MMD above global MMD, and final reconstruction no more than twice baseline.

Table 13: Complete MNIST entropy sweep. Median standard deviation is the effective posterior value; floor hit refers to the raw log-variance head.

| $\sigma_f$ | $H/\text{dim}$ | Med. std | Raw floor | Probe  | Global  | Cond.   | Gen-ACC | Rec.   |
|------------|----------------|----------|-----------|--------|---------|---------|---------|--------|
| 0          | -3.581         | .00674   | 100.00%   | 99.76% | .000579 | .016024 | 10.15%  | .01259 |
| .025       | -2.215         | .02589   | 99.16%    | 99.51% | .000599 | .016352 | 10.05%  | .01273 |
| .05        | -1.553         | .05045   | 99.17%    | 99.51% | .000648 | .016135 | 11.95%  | .01315 |
| .15        | -.477          | .15015   | 99.77%    | 98.78% | .000654 | .012044 | 37.35%  | .01520 |
| .35        | .381           | .35006   | 96.50%    | 95.61% | .000518 | .007516 | 93.50%  | .01658 |

HSIC attains  $p = 1/201$  at every level of the original MNIST seed-0 sweep and for every CIFAR-10 seed. This outcome is read alongside effect sizes: the three-seed CIFAR-10 .35 probe is only 10.5% and does not support practically recoverable label information even though the kernel test still detects dependence. Raw floor-hit is 100% at all CIFAR levels; the experiment manipulates effective noise rather than claiming that the variance head stops collapsing.

Table 14: Completed three-seed MNIST entropy confirmation. Probe is mean  $\pm$  sample standard deviation; the remaining entries are observed seed ranges supplied by the completed external runs.

| $\sigma_f$ | Probe (%)      | Global MMD             | Cond. MMD | Gen-ACC (%) |
|------------|----------------|------------------------|-----------|-------------|
| 0          | 99.5 $\pm$ .4  | $(5-7) \times 10^{-4}$ | .014-.018 | 9-13        |
| .15        | 97.5 $\pm$ 1.5 | $(6-8) \times 10^{-4}$ | .010-.014 | 32-42       |
| .35        | 93.0 $\pm$ 2.5 | $(4-6) \times 10^{-4}$ | .006-.009 | 85-96       |

Table 15: Seed-0 detail for the fixed CIFAR-10 entropy confirmation.

| $\sigma_f$ | $H/\dim$ | Med. std | Probe  | Global  | Cond.   | Gen-ACC | Rec.   |
|------------|----------|----------|--------|---------|---------|---------|--------|
| 0          | -3.581   | .00674   | 61.46% | .000249 | .005789 | 12.60%  | .05140 |
| .15        | -.477    | .15015   | 28.78% | .000743 | .003690 | 27.35%  | .05940 |
| .35        | .369     | .35006   | 9.76%  | .000667 | .001287 | 54.25%  | .06476 |

Table 16: Three-seed CIFAR-10 entropy confirmation. Entries are rounded means over model seeds;  $p = .00498$  means that every seed reaches the minimum plus-one calibrated HSIC value with 200 permutations.

| $\sigma_f$ | Probe (%) | Global MMD | Cond. MMD | HSIC $p$ | Gen-ACC (%) | Rec. L1 |
|------------|-----------|------------|-----------|----------|-------------|---------|
| 0          | 68.0      | .00030     | .0058     | .00498   | 13.0        | .052    |
| .15        | 29.0      | .00070     | .0037     | .00498   | 27.0        | .059    |
| .35        | 10.5      | .00067     | .0013     | .00498   | 52.0        | .065    |

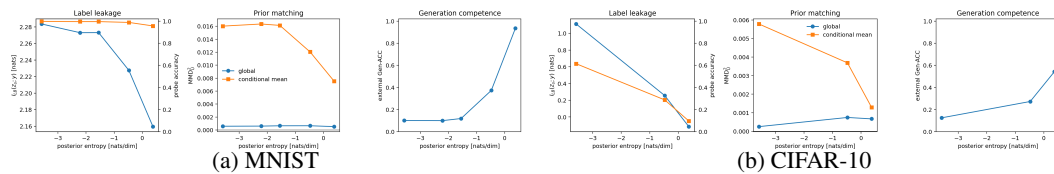


Figure 9: Leakage, marginal/conditional discrepancy, and independent Gen-ACC as effective posterior entropy increases. Curves are single-seed controlled interventions, not confidence intervals over training.

## H NATIVE CROSS-MODEL DETAILS

The fixed population contains 40,000 paired training examples and 10,000 paired test examples. Each canonical domain audit uses 1,000 balanced test examples. The independent Rotated-MNIST evaluator is trained only on real images and reaches 96.05% test accuracy. All models use their final 100-epoch checkpoint for seeds 0–2.

Table 17: Per-seed native cross-model results, macro-averaged over the two domains. Probe, content, and retention values are percentages.

| Family   | Seed | Global    | Cond.    | Logit | RBF   | HSIC domains | Content | Retention |
|----------|------|-----------|----------|-------|-------|--------------|---------|-----------|
| F-CS-WAE | 0    | .003287   | .032431  | 94.00 | 97.75 | 2/2 reject   | 99.25   | 12.55     |
| F-CS-WAE | 1    | .003209   | .030989  | 92.50 | 96.25 | 2/2 reject   | 99.50   | 11.10     |
| F-CS-WAE | 2    | .003481   | .033365  | 89.25 | 96.00 | 2/2 reject   | 99.50   | 13.45     |
| DRIT     | 0    | .259014   | .272479  | 25.00 | 25.50 | 2/2 reject   | 87.50   | 9.80      |
| DRIT     | 1    | .261287   | .274362  | 29.50 | 23.25 | 2/2 reject   | 90.00   | 10.00     |
| DRIT     | 2    | .245482   | .270487  | 48.75 | 48.75 | 2/2 reject   | 87.25   | 9.80      |
| DIVA     | 0    | .0000096  | .000268  | 11.50 | 12.50 | .0746, .0398 | 96.00   | 90.25     |
| DIVA     | 1    | -.0000452 | .0000288 | 11.00 | 11.25 | .2687, .0249 | 97.50   | 89.50     |
| DIVA     | 2    | -.0000181 | .000231  | 9.50  | 11.75 | .1194, .7264 | 96.50   | 88.60     |

All F-CS-WAE and DRIT domain-level HSIC values equal 1/201. For DIVA, Holm step-down correction over its six raw tests leaves no rejection. DIVA’s six global-MMD tests also do not reject. Its posterior entropy is approximately 1.36–1.38 nats/dimension; F-CS-WAE is approximately -3.58 and DRIT ranges from -1.88 to -1.57. The cross-model result therefore does not by itself remove posterior entropy as a confound for F-CS-WAE; that role belongs to Appendix G.

These are the diagnostics whose statistical object transfers without changing the native sampler. The direct term-3 and joint-replacement tests below are implemented for F-CS-WAE only; the table therefore supports cross-family discrimination by the shared subset, not a complete-contract claim for DRIT or DIVA.

## I POST-SPECIFIED COMPLETE-CONTRACT DIAGNOSTIC

This diagnostic was specified after the initial checkpoint audit and is therefore treated as post-specified. It does not retrain or select checkpoints: it loads the fixed F-CS-WAE seeds  $\{0, 1, 2\}$  and the same balanced Rotated-MNIST test populations used in Appendix H. Each domain supplies 100 posterior samples per digit. For every digit, a separate RNG stream draws 100 samples from the checkpoint’s Spherical Cauchy  $p(z_c | y = k)$  and 100 styles from  $\mathcal{N}(0, I)$ .

The term-3 test compares posterior and prior content with the chordal multiscale RBF kernel  $\sigma \in \{.1, .3, .5, 1, 2\}$ . The complete-contract test compares encoded pairs with independently sampled prior pairs using the product of that content kernel and the Euclidean style kernel with  $\sigma \in \{.5, 1, 2, 5, 10, 20, 50\}$ . Each reported domain statistic uses  $\sum_k \omega_k \widehat{\text{MMD}}_k^2$  with empirical  $\omega_k = n_k / \sum_j n_j$ . All audited domains contain 100 examples per digit, so this is numerically the mean of ten classwise unbiased MMD<sup>2</sup> statistics. Its 499 null replicates independently permute the two sample-bank labels inside each class before aggregation; plus-one calibration gives minimum  $p = 1/500$ . For each diagnostic, classwise decisions are corrected by Holm’s method separately within each of the six 10-digit seed–domain families.

Across all six seed–domain audits, term-3 MMD<sup>2</sup> is  $.00807 \pm .00347$  and direct-joint MMD<sup>2</sup> is  $.01943 \pm .00239$  (mean  $\pm$  sample standard deviation). Every aggregate test rejects. A single Bonferroni family over the 12 aggregate tests maps each raw  $p = .002$  to  $p_{\text{adj}} = .024$ , so all 12 conclusions remain. At the localized level, 56/60 term-3 and 60/60 direct-joint digit tests reject after Holm correction within each of the six 10-digit families for the corresponding diagnostic; these totals are not the result of one pooled 60-test correction. The direct result is an omnibus contract check, while the

Table 18: Held-out complete-contract diagnostics. All four aggregate tests within each row (two domains  $\times$  two tests) attain  $p = .002$ . Holm columns sum decisions from the two separate 10-digit domain families.

| Seed | $T_3^{-30}$ | $T_3^{+30}$ | $T_{\text{joint}}^{-30}$ | $T_{\text{joint}}^{+30}$ | $T_3$ Holm | Joint Holm |
|------|-------------|-------------|--------------------------|--------------------------|------------|------------|
| 0    | .004208     | .005682     | .016960                  | .017481                  | 16/20      | 20/20      |
| 1    | .006066     | .013177     | .017694                  | .022280                  | 20/20      | 20/20      |
| 2    | .008142     | .011161     | .020059                  | .022097                  | 20/20      | 20/20      |

term-3 result shows that content-prior mismatch contributes beyond condition leakage and within-condition dependence. The complete JSON, per-class null summaries, checkpoint/split hashes, and SHA-256 sidcar are stored under `runs_diag/complete_contract_diagnostics`; the result JSON hash is `aacbfd1456b02539...3c229838`.

## J SHAPES3D KNOWN-FACTOR AUDIT

Shapes3D provides 480,000  $64 \times 64$  images containing every combination of 10 floor hues, 10 wall hues, 10 object hues, 8 scales, 4 shapes, and 15 orientations. The fixed shape-stratified IID split contains 120,000 training, 20,000 validation, and 20,000 test images. F-CS-WAE and the learned C/S VAE use seeds  $\{0, 1, 2\}$ ; the observed-shape oracle has completed three-seed retention and factor-test follow-ups. Every model has 64 content and 64 style dimensions and is audited at epoch 100 on 4,096 examples (1,024 per shape). Global MMD uses 2,048 examples and 500 permutations; factor HSIC uses up to 1,024 examples and 200 permutations. Holm correction is applied separately within each latent view.

Table 19: Completed three-seed Shapes3D replication. Entries are rounded means over seeds  $\{0, 1, 2\}$ . The observed-shape oracle remains a seed-0 auxiliary control and is reported in the main paper.

| Model           | $z_s \rightarrow \text{shape}$ | $z_c \rightarrow \text{style}$ | Cond. shape MMD | Prior shape | Swap shape | Rec. L1 |
|-----------------|--------------------------------|--------------------------------|-----------------|-------------|------------|---------|
| F-CS-WAE        | 99.0                           | 87.0                           | .0085           | 89.0        | 88.0       | .0023   |
| Learned C/S VAE | 25.5                           | 9.5                            | .00004          | 99.7        | 99.8       | .0029   |

The completed observed-shape follow-up places  $z_c \rightarrow \text{style}$  in 9–13%, prior and swap shape retention in 99.5–100%, and raw style–shape HSIC  $p$ -values in .30–.80. Across seeds 1 and 2, the corrected pattern also repeats the seed-0 result: F-CS-WAE rejects all six factors in both views; the learned C/S VAE rejects only shape in content and the five nuisance factors in style; neither control rejects style–shape dependence after Holm correction. These are completed, manifest-verified follow-up results. The detailed table and plots below display the original seed-0 measurements where the retained aggregate export does not contain a seedwise numeric statistic; the corrected three-seed decisions and observed ranges are reported above without reconstructing unreported per-seed values.

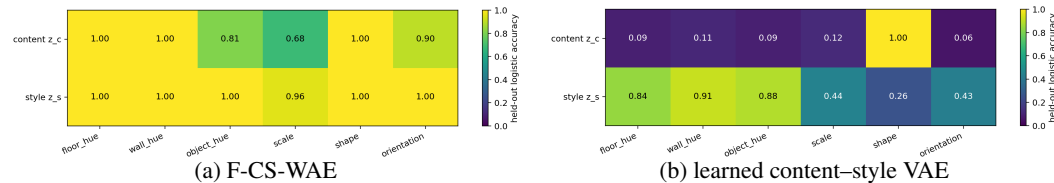


Figure 10: Held-out factor loading on Shapes3D stochastic latents. F-CS-WAE stores nearly every factor in both codes; the learned VAE places shape in content and nuisance factors in style. Accuracies are descriptive and are paired with the corrected dependence tests reported in the text.

After Holm correction, F-CS-WAE rejects dependence for all six factors in both views. Each control rejects only shape in content and only the five nuisance factors in style. The corresponding shape HSIC raw  $p$ -values in style are .597 for the observed-shape cVAE and .826 for the learned model.

Table 20: Seed-0 held-out logistic factor probes on stochastic latent samples. Columns follow dataset order. The intended pattern is content high only for shape and style low only for shape.

| Model               | View    | Floor  | Wall   | Object | Scale | Shape  | Orient. |
|---------------------|---------|--------|--------|--------|-------|--------|---------|
| F-CS-WAE            | content | 99.76  | 99.88  | 80.98  | 68.05 | 100.00 | 89.88   |
|                     | style   | 100.00 | 100.00 | 100.00 | 95.61 | 100.00 | 100.00  |
| Observed-shape cVAE | content | 9.76   | 9.51   | 9.02   | 14.27 | 100.00 | 8.29    |
|                     | style   | 100.00 | 99.63  | 99.02  | 45.49 | 25.98  | 39.76   |
| Learned C/S VAE     | content | 8.90   | 11.22  | 8.90   | 11.95 | 100.00 | 5.85    |
|                     | style   | 84.02  | 91.46  | 88.29  | 44.15 | 25.73  | 42.93   |

Table 21: Seed-0 independent factor-evaluator decoder checks. Posterior swaps keep content from the source and style from a different-shape donor; “target” is source shape for shape and donor value for nuisance factors.

| Model               | Floor  | Wall   | Object | Scale | Orient. | Swap shape | Prior shape |
|---------------------|--------|--------|--------|-------|---------|------------|-------------|
| F-CS-WAE            | 100.00 | 99.90  | 99.95  | 80.93 | 100.00  | 87.74      | 89.43       |
| Observed-shape cVAE | 99.98  | 100.00 | 99.80  | 95.02 | 99.93   | 100.00     | 99.88       |
| Learned C/S VAE     | 99.98  | 99.93  | 99.85  | 99.73 | 99.93   | 100.00     | 99.78       |

All three models pass the frozen competence checks: independent factor evaluator, content-shape probe, reconstruction L1 below .12, reconstruction shape accuracy above .8, and mean reconstructed nuisance-factor accuracy above .7. The observed-shape cVAE supplies content directly from the label; it is an oracle control for the intended split, not a like-for-like representation-learning competitor. The learned content–style VAE is the stronger negative control.

## K OFFICIAL SCFLOW EXTERNAL-VALIDITY AUDIT

### K.1 SCOPE GATE AND CLAIM BOUNDARY

This post-specified audit first applies a source-level scope gate rather than selecting a model from its numerical result. DEADiff exposes separate style and content branches, but its released inference path supplies the same reference image to both; independently assigning donors would therefore be our intervention rather than its declared sampling law. It is excluded. The official SCFlow implementation at commit `7c21af7dc3b3cfaff49876337b77b3aae6cfcdf5` passes the gate: forward inference accepts distinct content and style images, concatenates their CLIP embeddings, and training constructs the corresponding cross-image source. The released module also exposes forward merge and reverse endpoint recovery. We download the official SCFlow checkpoint and test embeddings, but not the separate UnCLIP decoder. Claims are consequently restricted to compatibility, transport, and retrieval in the released synthetic CLIP space; they do not cover decoded-image fidelity or diversity.

### K.2 FROZEN POPULATION AND TESTS

The official test split contains a complete 51-style by 3,000-content grid of 768-dimensional CLIP embeddings. For each evaluation-subset seed we select 20 contents and 20 styles, yielding 400 desired combinations. For each target  $(c, s)$ , the content donor has content  $c$  and a randomly selected different style, while the style donor has style  $s$  and a randomly selected different content. These donors instantiate the source law used by the official forward merger. Evaluation-subset seeds change only this held-out grid and donor assignment; they are not independently trained model seeds.

We retain the official inference default of one Euler step. Forward inference maps the concatenated donors to two target endpoints; their cosine agreement is above .994 in every subset, so we average them for target comparisons. Reverse inference duplicates the target embedding and recovers a content/style endpoint pair. Denote the recovered-pair law by  $Q^{\text{end}}$  and the law of the independently selected content/style donor pair by  $P^{\text{donor}}$ . We compare  $Q^{\text{end}}$  with  $P^{\text{donor}}$  using unbiased multiscale MMD and unbiased energy distance, each with 199 balanced pooled-label permutations. The same two tests separately assess both marginals. Delta-label HSIC tests whether recovered content varies with target style and recovered style with target content; Holm correction covers both directions across all three subsets. Transport reports forward target cosine and forward–reverse cycle cosine. Competence uses exact content, style, and joint nearest-neighbor retrieval against all 153,000 official test embeddings; simple donor averaging is the frozen baseline.

Table 22: Official SCFlow audit over three evaluation-subset seeds. Each row contains 400 combinations. Joint MMD and energy have  $p = 1/200$  in every row; both marginal tests do also. Cosine and joint retrieval show SCFlow/donor-mean baseline; cycle shows content/style.

| Subset seed   | Joint MMD | Joint energy | Target cosine | Cycle cosine | Joint ret. (%) | C/S ret. (%) |
|---------------|-----------|--------------|---------------|--------------|----------------|--------------|
| 2027          | .07002    | 2.8207       | .859/.765     | .251/.315    | 9.75/.00       | 17.75/71.50  |
| 2028          | .06585    | 2.6639       | .886/.792     | .249/.341    | 12.50/.25      | 18.75/81.00  |
| 2029          | .07158    | 2.8207       | .883/.787     | .236/.329    | 15.50/.00      | 19.75/79.50  |
| Mean $\pm$ SD | —         | —            | .876/.782     | .245/.328    | 12.58/.08      | 18.75/77.33  |

MMD and energy agree: joint, content-marginal, and style-marginal equality are rejected in all three subsets. Cross-factor dependence does not explain this failure. Only one of six directional HSIC tests has raw  $p < .05$  (.035), and none survives the familywise Holm correction. At the same time, the forward merger improves target cosine by  $.094 \pm .001$  and joint retrieval from  $.08 \pm .14\%$  to  $12.58 \pm 2.88\%$ . Style retrieval is stronger than content retrieval, while the low cycle cosines show that useful forward composition does not entail accurate inverse transport. This is the three-level distinction in one official checkpoint: endpoint-law compatibility fails, forward transport is useful but not invertibly consistent, and CLIP-space competence improves over a simple baseline. It is not a claim that SCFlow instantiates the encoder–prior equality in Theorem 1, that SCFlow is generically ineffective, or that exact equality is required for every downstream use.

The main summary is stored at `runs_modern/scflow_audit/summary.json` (SHA-256 `afc6330fc69d82c...4e3be60e02cbe41`). It embeds hashes for all three seed artifacts. The checkpoint hash is `51f3ed03013c0ed...37d8a265902054`; the official test-split hash is `b6d6b1338a2382ad...c514b5f0bb5c0c8`. The separate feasibility gate is `runs_modern/scflow_feasibility/gate.json`. A 100-combination pilot was used only to debug technical execution and is excluded from every aggregate.

## L CLAUSE-ALIGNED FACT INTERVENTION

### L.1 OBJECTIVE AND OPTIMIZATION

The intervention study uses the same MNIST architecture as the case study but a 40-epoch decision protocol with batch size 320. Phase boundaries are (5, 12, 26, 40); scheduled weights are frozen at zero-indexed epoch 34, and FACT becomes active at epoch 12. When active, the legacy semantic-class and per-class style penalties are replaced by the null-calibrated style, content, and sampled dependence constraints described in the following equation. For classwise statistic  $T_k$ ,

$$C_T = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \left[ T_k - \frac{1}{B} \sum_{b=1}^B T_{kb}^0 \right]_+, \quad B = 4, \quad (25)$$

where  $T^0$  uses matched prior–prior banks or within-class permutations. Reconstruction, aggregate content MMD, global style MMD, and semantic classification remain active. The direct joint statistic is never optimized.

Writing the three core positive-excess constraints as  $(C_s, C_c, C_d)$ , sampled style–label HSIC as  $C_y$ , posterior-mean conditional HSIC as  $C_\mu$ , and adversarial uniform confusion as  $C_a$ , the active

objective is

$$\mathcal{L}_{\text{FACT}} = \mathcal{L}_{\text{base}} + \lambda_s C_s + \lambda_c C_c + \lambda_d C_d + 100 C_y + 0.1 C_a + 4 C_\mu. \quad (26)$$

The adversary is a 128, 64 ReLU MLP, updated once per encoder minibatch with Adam learning rate  $10^{-3}$  and weight decay  $10^{-4}$ . Its classifier step uses detached sampled style; the encoder step freezes adversary parameters and minimizes  $\text{KL}(U||_{q_\phi}(y | z_s))$ . Both sampled and mean conditional-HSIC paths stop gradients into content.

The core multipliers start at one and are projected into  $[0, 10]$ . Once per epoch, the update for constraint  $j$  is

$$\lambda_j \leftarrow \Pi_{[0,10]} (\lambda_j + 0.1 \bar{C}_j / r_j), \quad (27)$$

with reference scales  $(r_s, r_c, r_d) = (.002, .0015, 2.5 \times 10^{-5})$  and an absolute per-epoch step cap of .25. All tolerances are zero after matched null subtraction; every null floor averages four independent draws.

For  $C_\mu$ , content means are L2-normalized and style means are standardized featurewise inside each represented class. The content kernel averages spherical chordal RBFs with bandwidths  $(.1, .3, .5, 1, 2)$ ; the style kernel averages Euclidean RBFs with bandwidths  $(.5, 1, 2, 5, 10, 20, 50)$ . Both Gram matrices are centered, their elementwise product is divided by  $(n_k - 1)^2$ , and the null permutes the standardized style Gram matrix within class. This is the same geometry as the held-out posterior-mean diagnostic.

## L.2 CALIBRATION, DEVELOPMENT PATH, AND FROZEN GATES

The retained development checkpoints form the cumulative path in Table 23. This is a sequential seed-0 development ablation: each row adds or corrects a component selected after the preceding row, so it explains the construction but is not a factorial causal estimate.

Table 23: Cumulative component path on development training seed 0 and audit RNG seed 0. Reductions and MLP are percentages; HSIC is relative to control.

| Cumulative configuration               | Joint ↓ | Cond. MMD ↓ | MLP   | HSIC/control |
|----------------------------------------|---------|-------------|-------|--------------|
| Schedule control                       | 0.00    | 0.00        | 72.68 | 1.000        |
| Core null-calibrated constraints       | 8.68    | 7.09        | 72.93 | 1.358        |
| + normalized estimator/dual correction | 21.62   | 30.85       | 56.59 | 1.437        |
| + sampled label HSIC                   | 30.73   | 21.53       | 48.29 | 1.572        |
| + nonlinear adversary                  | 30.33   | 28.97       | 11.46 | 1.069        |
| + posterior-mean HSIC ( $w_\mu = 4$ )  | 39.20   | 34.12       | 10.98 | .487         |

Calibration uses 20 MNIST minibatches from the frozen label-adversary checkpoint, with batch size 320 and four null draws. Mean raw conditional HSIC is  $.004647 \pm .000240$ , its matched-null floor is  $.003996 \pm .000328$ , and positive excess is  $.000651 \pm .000112$ . Target initial objective contributions of roughly .001, .003, and .01 yield the pre-specified weights  $\{2, 5, 20\}$ . Weight 4 is a final interpolation between the underpowered weight 2 and reconstruction-limited weight 5; it is selected on model seed 0 and then frozen before model seeds 1 and 2.

The seven gates are: semantic accuracy at least .985; reconstruction MAE no more than 1.10 times the schedule control; direct joint and conditional style MMD reductions of at least 20%; sampled-style MLP and RBF-SVM accuracies no greater than .40; and posterior-mean conditional-HSIC ratio no greater than 1.15. Evaluation seeds alter posterior/audit RNG and probe splits for the same model checkpoint; they are not independent training seeds.

Table 24: Seed-0 development path, averaged over evaluation seeds 0–2. Reductions, reconstruction change, and MLP are percentages. “Pass” counts evaluation seeds satisfying all seven gates.

| Run             | $w_\mu$ | Rec. $\Delta$ | Joint ↓ | Cond. ↓ | MLP   | HSIC/control | Pass |
|-----------------|---------|---------------|---------|---------|-------|--------------|------|
| Label adversary | 0       | 2.19          | 33.59   | 25.67   | 11.46 | 1.181        | 1/3  |
| Mean HSIC       | 2       | 3.45          | 35.26   | 26.72   | 11.06 | 1.196        | 1/3  |
| Mean HSIC       | 4       | 5.96          | 40.93   | 32.96   | 10.89 | .476         | 3/3  |
| Mean HSIC       | 5       | 10.49         | 44.27   | 29.04   | 12.85 | .966         | 0/3  |
| Mean HSIC       | 20      | 11.94         | 40.53   | 4.29    | 11.38 | .365         | 0/3  |

Weight 20 demonstrates that the new constraint can strongly lower its held-out target, but conditional style matching and reconstruction degrade. Weight 5 misses only the 10% reconstruction gate in all evaluation seeds. The interpolated weight 4 passes every gate for every evaluation seed and is the only configuration advanced to independent training-seed confirmation.

### L.3 DEVELOPMENT-ERA MATCHED CONTROLS

Training seed 0 is the development model. Seeds 1 and 2 are independently trained initial confirmations after all arguments are locked. We additionally train a schedule control for each of seeds 1 and 2, yielding matched control/FACT pairs for training seeds 0–2. All six checkpoints use audit RNG seed 0. Table 25 reports paired effects rather than differences from one fixed control.

Table 25: Matched control/FACT confirmation. Semantic  $\Delta$  is in percentage points; reconstruction and reductions are relative percentages. MLP shows control  $\rightarrow$  FACT. HSIC  $\downarrow < 0$  denotes a worsening. Every training seed is retained.

| Training seed | Semantic $\Delta$ | Rec. $\Delta$ | Joint $\downarrow$ | Cond. $\downarrow$ | MLP control $\rightarrow$ FACT | HSIC $\downarrow$ | Gates     |
|---------------|-------------------|---------------|--------------------|--------------------|--------------------------------|-------------------|-----------|
| 0             | −.45              | +5.94         | 39.20              | 34.12              | 72.68 $\rightarrow$ 10.98      | 51.34             | 7/7       |
| 1             | −.50              | +12.35        | 35.09              | 13.92              | 80.49 $\rightarrow$ 11.46      | −30.89            | 4/7       |
| 2             | + .10             | +1.26         | 42.37              | 28.39              | 77.56 $\rightarrow$ 10.98      | 36.59             | 7/7       |
| Mean          | −.28              | +6.52         | 38.89              | 25.48              | 76.91 $\rightarrow$ 11.14      | 19.01             | 2/3 seeds |
| SD            | .33               | 5.57          | 3.65               | 10.41              | —                              | 43.84             | —         |

Absolute joint MMD is  $.002368 \pm .000160$  for controls and  $.001449 \pm .000166$  for FACT; absolute classwise style–prior MMD is  $.002567 \pm .000153$  and  $.001902 \pm .000148$ , respectively. Every aggregate joint test rejects at  $p = .005$  before and after the intervention, so the paired reduction does not establish exact compatibility. Seed 1 also fails the reconstruction, conditional-style-MMD, and posterior-mean-HSIC gates. The nonlinear leakage effect is stable, whereas mean conditional dependence is not.

### L.4 PRELIMINARY POST-FREEZE TRAINING SEEDS

After the entire development and initial-confirmation sequence, training seeds 3–5 are run from epoch 1 without changing any argument. These are clean post-freeze checks and are not pooled with the matched seed 0–2 analysis. Their pre-specified gates retain the frozen schedule control seed 0 as the reference, exactly as declared before the runs.

Table 26: Complete post-freeze confirmation. Reduction and MLP columns are percentages; HSIC is a ratio to the frozen schedule-control seed 0.

| Training seed | Semantic | Rec. MAE | Joint $\downarrow$ | Cond. $\downarrow$ | MLP   | HSIC/control | Gates     |
|---------------|----------|----------|--------------------|--------------------|-------|--------------|-----------|
| 3             | 98.80    | .020477  | 22.53              | 20.36              | 11.46 | .373         | 7/7       |
| 4             | 99.20    | .021081  | 18.02              | 12.39              | 11.46 | .583         | 5/7       |
| 5             | 98.95    | .021055  | 29.43              | 25.83              | 11.46 | .624         | 7/7       |
| Mean          | 98.98    | .020871  | 23.33              | 19.53              | 11.46 | .527         | 2/3 seeds |
| SD            | .20      | .000342  | 5.75               | 6.75               | .00   | .135         | —         |

Seed 4 misses only the two 20% distributional-reduction gates. It passes semantic competence, reconstruction, both nonlinear probes, and the posterior-mean-HSIC gate. These preliminary outcomes motivate reporting the matched seed-level rows, including their adverse results, as the intervention evidence in the main paper.

### L.5 POST-SPECIFIED KERNEL-FREE ROBUSTNESS

We rerun the frozen sample-bank audit with unbiased energy distance in the declared latent coordinates and replace conditional HSIC by distance correlation. Content uses chordal distance on the unit sphere; style uses the native unit-Gaussian-prior scale; the joint concatenates  $z_c/\sqrt{2}$  and

$z_s/\sqrt{2d_s}$ . Each statistic uses 200 samples per class per bank and 199 matched permutations. This analysis was specified after the MMD/HSIC results and is not used for model selection.

Table 27: Kernel-free robustness. Reductions are percentages relative to the matched control (seeds 0–2) or frozen control seed 0 (seeds 3–5); positive values favor FACT. “Agree” is MMD–energy equality-decision agreement over 33 aggregate/classwise comparisons.

| Role        | Seed | Joint MMD ↓ | Joint energy ↓ | Cond. dCor ↓ | Agree  |
|-------------|------|-------------|----------------|--------------|--------|
| Matched     | 0    | 39.20       | −204.07        | 38.46        | 96.97  |
| Matched     | 1    | 35.09       | 48.49          | 7.11         | 100.00 |
| Matched     | 2    | 42.37       | 55.37          | 5.73         | 100.00 |
| Post-freeze | 3    | 22.53       | 31.13          | 24.98        | 96.97  |
| Post-freeze | 4    | 18.02       | −15.77         | 37.90        | 96.97  |
| Post-freeze | 5    | 29.43       | −108.70        | 32.44        | 100.00 |

Both dependence estimators reject within-class independence for every model, while HSIC and distance correlation decrease for all three matched pairs and all three preliminary post-freeze models. MMD and energy agree on 99.0% of the equality-test decisions, showing that contract rejection is not specific to the RBF kernel. This decision agreement must not be read as agreement about intervention magnitude: joint energy improves in only two of three matched pairs and one of three post-freeze comparisons, with large adverse values for matched seed 0 and post-freeze seed 5. The main paper therefore limits the intervention claim to the selected MMD diagnostic and decoder attribution.

## L.6 PAIRED DECODER INTERVENTIONS

The independent pixel classifier evaluates deterministic posterior-mean swaps and 10,000 prior samples per checkpoint. The class-conditional style sampler is a post-hoc diagnostic: it deliberately exploits label-bearing style and is not the declared factorized generation law.

Table 28: Paired decoder effects. Swap columns are FACT minus control in percentage points. Gen-ACC shows control → FACT.

| Training seed | Swap-content Δ | Swap-style Δ | Global Gen-ACC | Class-style Gen-ACC |
|---------------|----------------|--------------|----------------|---------------------|
| 0             | +21.00         | −25.70       | 38.52 → 26.85  | 99.74 → 11.18       |
| 1             | +34.70         | −35.22       | 38.57 → 31.92  | 99.96 → 96.93       |
| 2             | +20.53         | −26.33       | 35.37 → 23.90  | 99.68 → 28.45       |
| Mean Δ        | +25.41         | −29.09       | −9.93          | −54.27              |
| SD            | 8.05           | 5.32         | 2.84           | 45.22               |

All swap pairs move in the intended direction, directly showing less decoder reliance on style. All global-prior pairs lose Gen-ACC. FACT therefore improves an audited latent relation and a posterior-swap mechanism, but does not restore the intended prior-generation behavior. The high variance of the class-style diagnostic further shows that its shortcut is not a stable target.

## L.7 INTERVENTION ARTIFACTS

Calibration is stored at `runs_diag/fact/label_adversary_mnist_40e/mean_hsic_calibration.json` (SHA-256 776540a7ebed3a3aa...42eeb23). The evaluation-seed summary is under `runs_diag/fact/mean_hsic_mnist_40e/bridge_w4` (SHA-256 65698f6aaa346409...6bbf79); the independent model-seed summary is under `runs_diag/fact/mean_hsic_mnist_40e/model_seed_confirmation_w4` (SHA-256 fe69797ae30deba7...b704705). The matched-control, post-freeze, and decoder-intervention summary is `runs_diag/fact/mean_hsic_mnist_40e/submission_followup_v1/summary.json` (SHA-256 6b1e693862820af9...36f73d0cb44); its manifest hash is `f5eea2651fc76aad...f7ald5556b77`. It records source hashes, exact commands, checkpoint hashes, training-seed roles, audit RNG seed, and every adverse result. Checkpoints are saved every epoch with RNG, optimizer, scheduler, dual, and adversary state; resume rejects configuration mismatches. The implementation is tested by 27 focused loss, gradient, dual-update, phase, adversary, and checkpoint round-trip tests. The

kernel-free audit is stored at `runs_diag/kernel_robustness/summary.json` (SHA-256 `11d2e5358023fa14...159fca4f`); it records both protocol-input hashes and every matched/post-freeze value.

## M REPRODUCIBILITY AND EVIDENCE BOUNDARY

The original checkpoint audit uses protocol `stage0-1.1.0`; synthetic, entropy, cross-model, and Shapes3D audits use `stage0-1.2.0`; the kernel-free extension uses `stage0-1.3.0`; the official SCFlow audit uses `scflow-contract-audit-1.1.0`. Table 29 collects the primary roots rather than scattering paths through the main text.

Table 29: Primary evidence roots in the anonymous artifact snapshot.

| Evidence family    | Relative artifact root                                |
|--------------------|-------------------------------------------------------|
| Exact marginal     | <code>runs_diag/synthetic_exact_marginal_paper</code> |
| Entropy            | <code>runs_f/entropy_sweep</code>                     |
| Cross-family       | <code>runs_cross_model/native_v1</code>               |
| Shapes3D           | <code>runs_shapes3d/factor_audit_v1</code>            |
| Complete contract  | <code>runs_diag/complete_contract_diagnostics</code>  |
| FACT intervention  | <code>runs_diag/fact/mean_hsic_mnist_40e</code>       |
| Kernel-free checks | <code>runs_diag/kernel_robustness</code>              |
| Official SCFlow    | <code>runs_modern/scflow_audit</code>                 |

Rounded follow-up values are consolidated in `fcsvae/reported_followup_aggregates_v16.json`. Its SHA-256 sidecar fixes the aggregate export at `83e4c9038f944782...60aa665ca1d63d4`. Each final JSON has a SHA-256 sidecar or embeds the criteria, checkpoint, split, evaluator, source-file, and repository hashes. Synthetic criteria are frozen in `acceptance_criteria.json`, with the composite decision in `acceptance_result.json`. Entropy summaries record the frozen CIFAR promotion decision. Cross-model reporting is derived from nine native checkpoint audits without using the legacy proxy artifacts. Shapes3D suite completion records all three models as complete and competence-passing. The complete-contract artifact records its post-specified status, independent sample banks, kernel ladders, permutation counts, and checkpoint/split hashes. The completed sample-size sweep and additional three-seed MNIST, Fashion-MNIST, CIFAR-10 entropy, and Shapes3D runs are reported here as rounded aggregate exports. Each follow-up was run under the frozen audit protocol and checked against its manifest and SHA-256 sidecar before aggregation; the corresponding run artifacts are retained in the audit archive separately from this source repository. We preserve the verified mean/SD or observed-range semantics and do not reconstruct unreported standard deviations or aggregate  $p$ -values from cross-experiment CV bands. Seedwise decision claims are stated only when the calibrated decision agrees across every recorded run.

Some historical artifacts record a dirty worktree; that provenance is retained rather than relabeled. The anonymous snapshot includes the exact critical source files and `fcsvae/v26_evidence_manifest.json`, whose generated sidecar binds them to the traceable headline summaries. The present scope is 50 synthetic replications at the primary  $n = 2,048$  setting, four sample sizes at  $\kappa = 16$ , three training seeds per native cross-model family, three-seed original-image headlines, three CIFAR-10 entropy seeds, three F-CS-WAE/learned-VAE Shapes3D seeds, three MNIST entropy seeds, three observed-shape-oracle follow-up seeds, three matched MNIST control/FACT pairs, and three additional post-freeze FACT training seeds. The official SCFlow result uses one released checkpoint and three seeded 400-combination evaluation subsets; these are evaluation seeds, not model seeds. Aggregates without locally available seed-level artifacts are excluded from this version. We make no uncertainty claim beyond those boundaries. Submission source, rendered author blocks, and supplementary metadata must remain anonymous during review.